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Radar Gait Recognition Using Attention-based Dual-stream Vision Transformer

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Abstract

Recent years have witnessed various approaches of gait recognition and its applications. Compared to the traditional image-based gait recognition, radar gait recognition is robust to light variations and less infringement on privacy. Previous studies often utilise either spectrograms or cadence velocity diagrams. While the former shows the time-frequency patterns, the latter encodes the repetitive frequency patterns. In this work, a gait recognition mechanism using radar micro-Doppler signature has been developed. Additionally, a dual-stream neural network with attention-based fusion is proposed to fully aggregate the discriminant information from these two representations. The both streams are designed based on the Vision Transformer, which well captures the gait characteristics embedded in these representations. The proposed method is validated on a large benchmark dataset for radar gait recognition, which shows that it significantly outperforms state-of-the-art solutions.

Publications

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List of Abbreviations

ADS-ViT	attention-based dual-stream Vision Transformer
CNN	convolutional neural network
CVD	cadence velocity diagram
DCNN	deep convolutional neural network
FFT	fast Fourier transform
GEI	gait energy image
GPU	graphics processing unit
ICASSP	IEEE International Conference on Acoustics, Speech and Signal Processing
IQ	in-phase/quadrature
KNN	k -nearest-neighbour
LSTM	Long Short Term Memory
mDS	micro-Doppler signature
MLP	multi-layer perceptron
PPE	Position and Patch Embedding
STFT	short-time Fourier transformed
SVM	support vector machine
TCN	temporal convolutional network
TE	Transformer Encoder
UNNC	University of Nottingham Ningbo China
ViT	Vision Transformer

Chapter 1

Introduction

This report describes a novel and high-performance method developed by authors for radar gait recognition. **The work has been successfully accepted by a CCF-B conference, IEEE International Conference on Acoustics, Speech and Signal Processing.** In this chapter, a brief introduction to the final year project is presented. Section 1.1 and 1.2 give the background and motivation of the project. Section 1.3 lists and explains the aims and objectives of this project. Section 1.4 illustrates the overall process of conducting this study. Section 1.5 briefly summarises our contributions to the specific research area of radar gait recognition. Section 1.6 outlines the thesis structure.

1.1 Background

Security and privacy are two major concerns of people when considering organisational secrecy and personal information. Several biometric identifiers such as fingerprint, face recognition and iris recognition have long been used to authenticate at short or even close distances. Gait, the pattern of walking, is a biological characteristic that can be measured at great distances because of its property to be relatively macro. It is therefore a great supplement approach to the aforementioned biometrics in some specific situations where short-distance identification is unavailable. Additionally, gait recognition has become an attractive research area with many applications such as public safety monitoring [11], health screening [12] and human-computer interaction [13, 14].

A traditional way of representing walking people is using sequences of images captured by optical sensors [1, 15]. However, this approach has two major drawbacks. First,

since gait features only vary slightly from the front view, this approach requires the images to be captured from the side. This highly reduces the practicality as multiple cameras are necessary for setting up the system. Second, optical sensors depend heavily on lighting conditions. This means the system may function well in daytime but can fail with the inference of different weather conditions and strong light. This further limits its application scenarios.

Radar is a more advanced tool for recording gait characteristics, which addresses the front view issue and is robust to lighting conditions. It is commonly used to detect targets and measure the distance, angle or velocity of objects by transmitting electromagnetic waves and receiving the echo. Doppler effect, the waves frequency change caused by the relevant movement between the target and the radar, is one important measure that radar can obtain. The micro-Doppler effect refers to the frequency change of the target's components induced by the micro-motion of them. Doppler radar can capture the micro-Doppler signatures (mDS) of a moving target's gait feature from the front [10], and it has been demonstrated that mDS can reveal human dynamic features and ascertain human actions and gestures [10, 16, 17]. More importantly, the radar signal does not catch images of targets so is less invasive of privacy and works even in low-light conditions, making it robust in a variety of real-world situations [14]. Thus, in this project, we mainly focus on recognition using mDS. An illustration of recognition using mDS is shown in Figure 1.1.

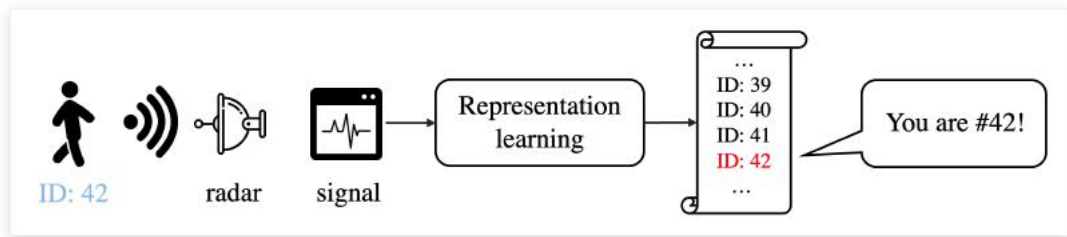


Figure 1.1: An illustration of recognition using mDS.

1.2 Motivation

Many mDS representations have been developed, *e.g.*, spectrograms [18–20], cadence velocity diagrams (CVD) [21] and cepstrograms [22]. The spectrogram is a time-varying representation of mDS in the joint time-frequency domain [10]. As the micro-Doppler effect induced by mechanical vibrating or rotating structures of a target is stable and constant, the Doppler frequency caused by that target could be well formulated by a function of dwell time [23]. Thus, the time-frequency spectrogram of mDS could be useful to represent a moving target and further be commonly used in gait recognition and classification [17, 24–27]. However, even the same person could have slightly different gait cycles and different start locations of each cycle, which would lead to degradation in classification performance. The repetition information of those frequencies is scarcely considered but is also the invariant feature of gait cycles.

The Cadence Velocity Diagram is another form of mDS converted from the spectrogram by taking Fast Fourier Transform (FFT) along the time axis [21]. The CVD is less commonly used in the gait recognition [28, 29]. The CVD provides another useful measure on the frequency of different velocities of body parts repeated, which is complementary to the spectrogram.

Different kinds of traditional learning paradigms, such as Naïve Bayes [30], support vector machines [12] and k -nearest-neighbour classifiers [31] have been used to classify mDS. Several researchers have investigated deep convolutional neural networks (CNN) to implement the classification of human gestures and human actions [9, 26]. [10] has developed a temporal convolutional neural network for spectrogram, and [32] has utilised a dual-channel network to classify gait features. However, they did not exploit the latent of CVD.

In this work, an attention-based dual-stream Vision Transformer is proposed to identify features that are not hand-crafted and further improve classification performance.

Since the time-frequency representation of mDS only contains the time-varying nature of Doppler frequencies, CVD is introduced to provide complementary information on how often different frequencies repeat. Furthermore, an attention-based fusion is applied to fuse the most useful features. To our knowledge, this is the first investigation in fusing the two discriminating clues. We also constructed a dual-stream neural network to extract features from these two representations respectively. Moreover, we first employ Vision Transformer as the backbone of our neural network, utilising the patching function of it to deeply exploit the physical meanings of both representations, and achieve an accuracy of 91.2% in a 98-people dataset.

1.3 Aims and Objectives

The main objective of this project is to develop a deep-learning-based method in order to recognise people by their radar gait features in multiple representations. The method would be able to learn individuals' gait features through several well-processed radar time-frequency diagrams and recognise their identity by the learned model using deep learning. Apart from the main objective, the method is to be presented at a conference. The main objective is divided into several sub-aspects to achieve:

1. Study the traditional gait recognition.
2. Study the micro-Doppler effect and justify the possibility of applying it to recognise human gait.
3. Process the raw radar signal for clear, recognisable and regular diagrams. The raw micro-Doppler radar signal is obtained in a 2D representation, and this step is to investigate the signal processing techniques to obtain suitable diagrams for learning.
4. Further study different mDS representations and explore their practicalities and functionalities in representing gait features of walking people.

5. Explore and investigate possible deep learning methods existing in the industry to test and find an efficient way to recognise gait features.
6. Design and develop a novel method and apply it to obtain better performance in gait recognition.
7. Test and improve the method and model.
8. Compare and evaluate the performance of existing methods with the method we proposed.

1.4 Study Design

The spectrogram is commonly used to represent mDS, while the CVD is often ignored though it provides additional complementary knowledge to the spectrogram. In our study, we defined a novel universal dual-stream structure for extracting both features to improve the accuracy of gait recognition. Figure 1.2 shows the overall procedures of this study. We first focus on signal processing, since the raw Doppler radar signal is in a 2-dimension form which consists of background noise and long in time scale. The produced spectrogram will be utilised and additionally converted to CVD for another stream. A dual-stream neural network is then responsible for extracting features from these two representations. The extracted features are fed into Attention-based fusion to aggregate for final classification. The results of recognition are evaluated and used to improve the architecture and other detailed designs, which forms the iteration of the study.

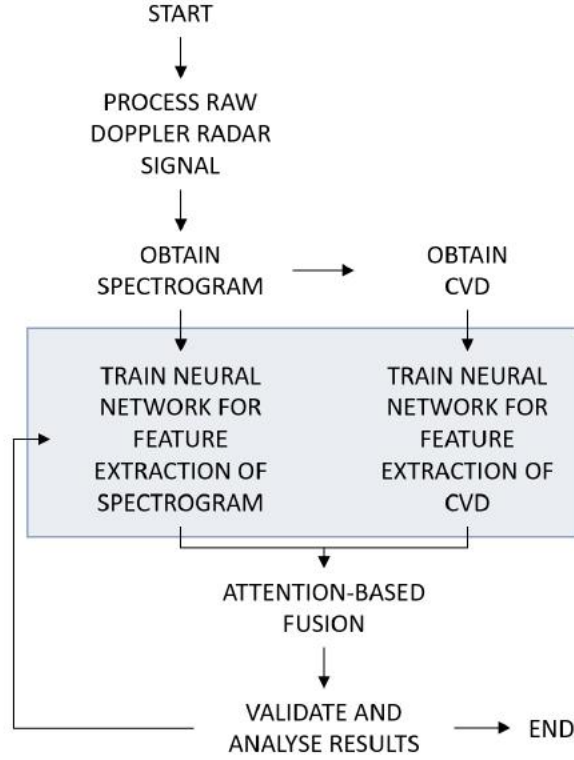


Figure 1.2: An overview of the study.

1.5 Contributions

1. We use radar micro-Doppler signatures to represent human gait characteristics instead of images captured by optical sensors. This addresses the issues of front-view recognition and light variations.
2. We develop an attention-based dual-stream Vision Transformer (ADS-ViT) which uses the Vision Transformer as the backbone for the first time in the field of radar signal classification. The patch mechanism of ViT helps learn physical information embedded in mDS and obtain leading performance.
3. We introduce the use of an attention-based fusion scheme, which learns and fuses useful features from two mDS representations and significantly improves the performance.

1.6 Thesis Structure

The remainder of the report is structured as follows. Relative literature is reviewed and explained in **chapter 2**. **Chapter 3** demonstrates the proposed method of the attention-based dual-stream Vision Transformer in radar gait recognition and explains why the dual-stream architecture with Vision Transformer would be a better choice. **Chapter 4** shows the experimental results with analysis, illustrating the performance of our approach. **Chapter 5** summarises the work and gives reflection.

Chapter 2

Literature Review

In this chapter, a brief introduction to the traditional gait recognition method using images captured by optical sensors is given in Section 1. Section 2 explains the micro-Doppler effect and signatures. Section 3 further describes two representations of mDS. Section 4 introduces two types of fusion schemes. Section 5 contains reviews of several related works that use Spectrograms to represent gait features.

2.1 Image-based Gait Recognition

In literature, gait cycles are commonly captured by cameras' optical sensors [1, 2, 15]. Two main approaches are widely used in literature. The first approach initially processes sequences of gait images into silhouettes (as shown in Figure 2.1) to drop all unrelated information such as the environment and the colours, keeping only the shapes of bodies. Gait silhouettes are then treated as one image or a gait template [2]. Another approach, however, is model-based and utilises the original images directly in order to retain temporal and spatial information as much as possible [1].



Figure 2.1: Example silhouettes of several gait cycles of a person [1].

One typical example of treating a gait sequence as one image is Gait Energy Image (GEI) [2]. GEI is a spatio-temporal gait representation, representing the human gait sequence in one single image. The original silhouettes sequence is first sliced into patches of the same size, and the body shapes are placed at the centre of each patch. Then all the patches are superimposed into one image by averaging every pixel value. In this way, a GEI is established with spatial information represented in the magnitude of pixel values. An example of establishing a GEI can be viewed in Figure 2.2. Because of the averaging operation, this approach is less sensitive to noise induced by some gait frames. However, GEI highly depends on the camera’s view angle and frame rate and is not able to deal with clothing variations. Some research has focused on improving the robustness in certain variations. The camera’s view angle issue has been discussed in [33], proposing to transform gait images from any angle to a specific required one. The results were still not promising as temporal information was dropped [15].



Figure 2.2: Example establishment of a Gait Energy Image [2].

The whole original sequence of gait images is usually processed with neural networks to extract human body structure for recognition. 3D convolutional neural networks have been directly applied to the sequence of gait images in [34] to capture both spatial and temporal features of gait. In addition to better performance in standard gait recognition, the system can resolve variations in clothing and large view angle change. In [35], the authors utilise Long Short Term Memory (LSTM) and Convolutional Neural Networks (CNN) to record temporal memory and further extract pose information. This helps improve the robustness in carrying variations while still keeping necessary spatial and

temporal information. It is also suggested by [36] that identifying humans by body joints from Kinect is already promising, but manual marking and huge computational costs are still the issues.

In conclusion, such optical-sensor-based gait recognition systems may face insurmountable challenges, such as extracting explicit human silhouettes from a complex environment background, dealing with silhouette deformation caused by partial occlusions by carrying variations, and dealing with frontal-view gait analysis [37].

2.2 Micro-Doppler Signatures

Radar, which refers to radio detection and ranging, is originally used to detect objects, measure the distance between and obtain the target's speed. The distance is calculated by the time delay of the emitting radio and the echo received if the target is relatively stationary with the radar. However, the relative motion will induce the frequency change of the radio wave received, and this is also known as Doppler effect [21]. More specifically, the received wave frequency becomes higher if the target is moving towards the radar while lower when it is moving away. An illustration of the Doppler effect can be viewed in Figure 2.3.

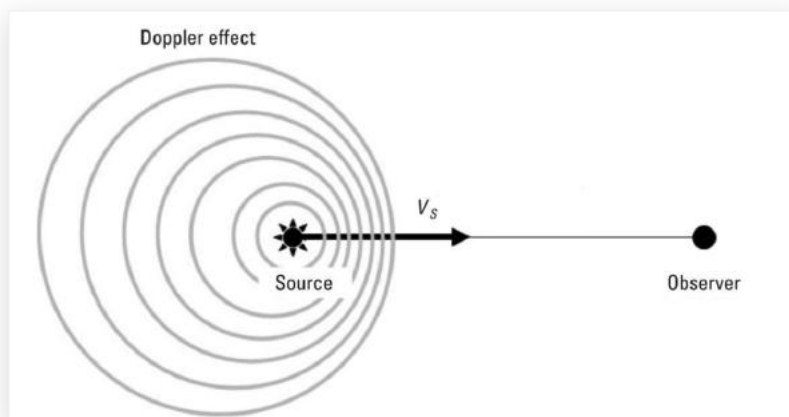


Figure 2.3: The source and the observer is approaching each other. The waves between are compressed, and the frequency is therefore higher. Adopted from [3].

The above is the Doppler effect presented in a macroscope, while any small structural components of a target can also generate relative motion and thus induce micro-Doppler signatures, *i.e.* each part has its own Doppler signature [3]. Walking people is a clear example of how micro-Doppler exists. A walking person has several relatively moving components, *e.g.* swinging legs and arms and torso. Each moving part of a person can generate different micro-Doppler signatures depending on personal habits, which can be unique identification for individuals.

2.3 Representations of micro-Doppler Signatures

The raw data generated by Doppler radar is simply represented in a two-dimensional form, *i.e.*, the magnitude of the echo signal in real numbers and complex numbers. This form of representation, although contains most information, is hard to read and explain. The method of time-frequency analysis is introduced to process the raw radar signal to exploit the time-varying latent of mDS. Several time-frequency analysis has been proposed in the literature, *e.g.* spectrograms, cadence velocity diagrams and cepstrograms. In this section, two complementary representations of mDS are introduced. Section 3.3.1 describes a time-frequency representation of micro-Doppler signatures. Section 3.3.2 introduces the cadence velocity diagram derived from the spectrogram and the extra knowledge it provides.

2.3.1 Spectrogram

A walking person normally has their arms and legs swinging, as well as their body's speed forward changing. Micro-Doppler signatures are induced by the micro-motion of these body parts. The varying speed of body parts corresponds to the frequency changes over time in radar signals and are well recorded in mDS. Tahmoush and Silvius [38] modelled Doppler of each body part of a walking human as sinusoidal modulation in the spectrogram. This model reveals the velocities of each body part and determines the

necessary human gait characteristics. A sample for illustration can be seen in Figure 2.4.

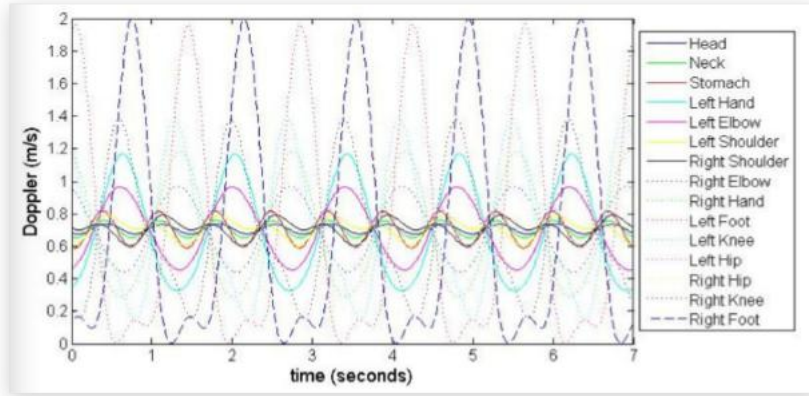


Figure 2.4: Sample illustration of sinusoidal modulation of body parts in the spectrogram.

The extracted features are the mean Doppler velocity and the size of torsos in the spectrogram. The mDS of a person is found relatively consistent and different walking people exhibit discriminative characteristics in the spectrogram [38]. The radar spectrogram can well manifest the time-varying characteristics of a person’s gait. A real spectrogram from our dataset is shown in Fig. 2.5. This figure records a person walking towards the radar with most sinusoidal curves above 0 Hz. The sinusoidal curves with the highest amplitude depict the swing of legs, as legs have the largest relative speed change towards radar among all the body parts. The sinusoidal amplitude for arms is around 40 Hz, and the 0 Hz middle curve stands for the torso which keeps a relatively stable speed while walking.

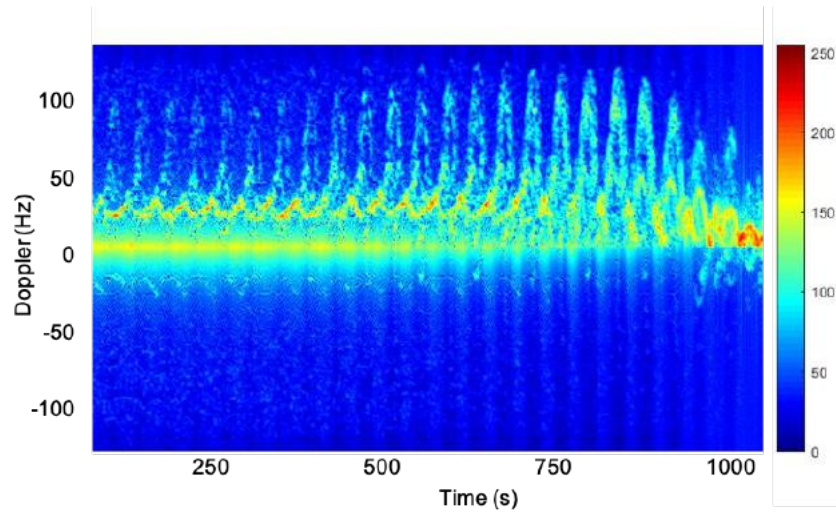


Figure 2.5: Example gait mDS spectrogram representing a person walking towards the radar.

Gait signatures can be well recorded even when a person is walking backwards. This is due to the symmetric nature of the Doppler effect. As Figure 2.6 shows, a person walking away from the radar simply generates a spectrogram with sinusoidal curves under 0 Hz, and this can be easily transformed to fit the previous orientation for recognition.

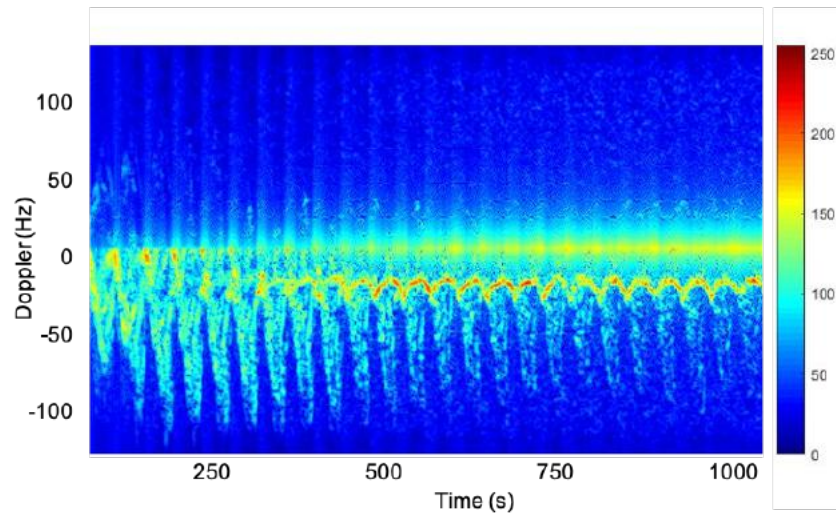


Figure 2.6: Example gait mDS spectrogram representing a person walking away from the radar.

2.3.2 Cadence Velocity Diagram

The spectrogram only provides frequency distribution at a certain time frame, while the frequency of repetitions of velocities is somehow missed. After obtaining the spectrogram, the CVD [21] can be derived by taking the Fourier transform of the spectrogram along the time axis. The derived CVD, as shown in Fig. 2.7, is a matrix with rows representing Doppler frequencies and columns representing cadence frequencies, which measures how frequently different frequencies appear in the signal over the observation duration. It encodes useful information on the repetition of velocities, which would be the key feature to identifying a walking person. As a result, the CVD provides data on the shape, size, and frequency of the curves in the original spectrogram.

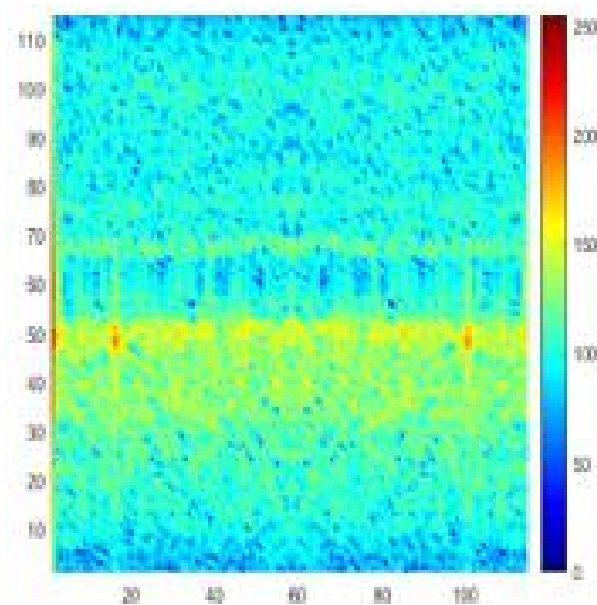


Figure 2.7: Example cadence velocity diagram.

2.4 Decision and Feature Fusion

Data size, dimension of the feature space and types of features can vary greatly for pattern recognition jobs. These variations could considerably affect the classification performance. Most modern ways of tackling these issues are using multiple classifiers or multiple feature representations to improve the performance. Fusion is a common practice

in order to aggregate different results from multiple classifiers or different features. Fusion can therefore be categorised into decision fusion and feature fusion [39].

Decision fusion, also known as score-level fusion, only considers the probability vectors computed by multiple classifiers, such as support vector machines [12] and k -nearest-neighbour classifiers [31]. Different decisions made by these classifiers are then merged using various schemes to obtain final classification results. Research has investigated and shown the effectiveness of decision fusion [40]. Feature-level fusion, which aims to utilise all or part of features from different sources, can be achieved by concatenation, averaging, max pooling, min pooling, etc [39]. The fused features are fed into a classifier for the final decision. Attention-based feature-level fusion is another approach that could memorise the relationship and decide the importance and confidence of both features and therefore provide more promising results [41].

2.5 Deep Learning Methods

Machine learning has recently been a great success due to the great improvement in the hash rate of the graphics processing unit (GPU). Deep learning is a machine learning method with many layers to be more complicated to extract useful features without human intervention by learning weights over epochs [34]. Three dedicated designed deep learning architectures are introduced in the following of this section.

2.5.1 AlexNet

AlexNet is a deep convolutional neural network (DCNN) first introduced to do classification tasks on ImageNet [4]. As a deep-learning-based approach, AlexNet is able to learn features directly from input to perform end-to-end classification. The performance is enhanced through deeper convolutional layers extracting different levels of features. AlexNet contains five convolutional layers, three max-pooling layers, two normalization layers, two fully connected layers and one softmax layer [4]. An illustration of AlexNet

can be viewed in Figure 2.8. It has been applied by [9] to perform the classification of mDS.

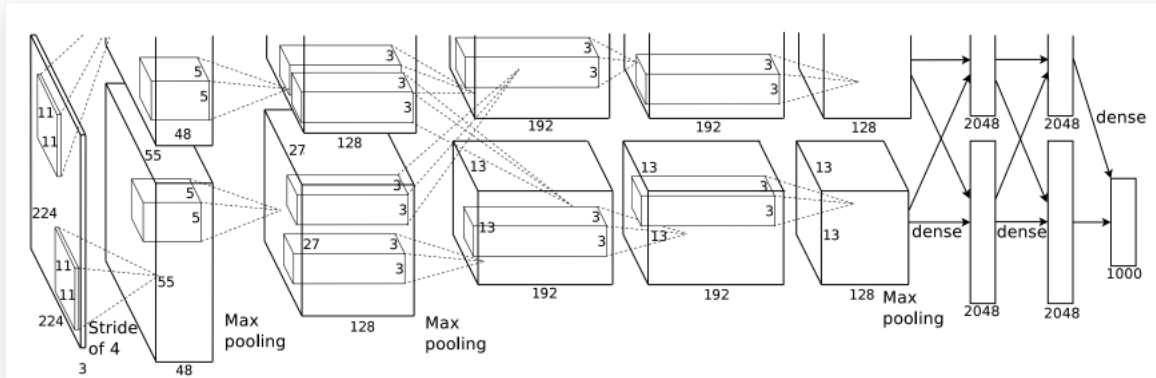


Figure 2.8: Architecture of AlexNet, adopted from [4].

2.5.2 VGG

VGG is later introduced by [5] with the idea that deeper is better. It extends the number of convolutional layers from five of AlexNet to thirteen for a standard VGG-16 network, and it has three more fully connected layers and two Dropout layers which further mitigate the overfit issue induced by more layers. The fundamental intuition is that more layers can identify more sophisticated textures as well as general shapes of objects. It has long been considered baseline in classification tasks and is adopted to recognise radar signatures as well [10]. The architecture of VGG is shown in Figure 2.9.

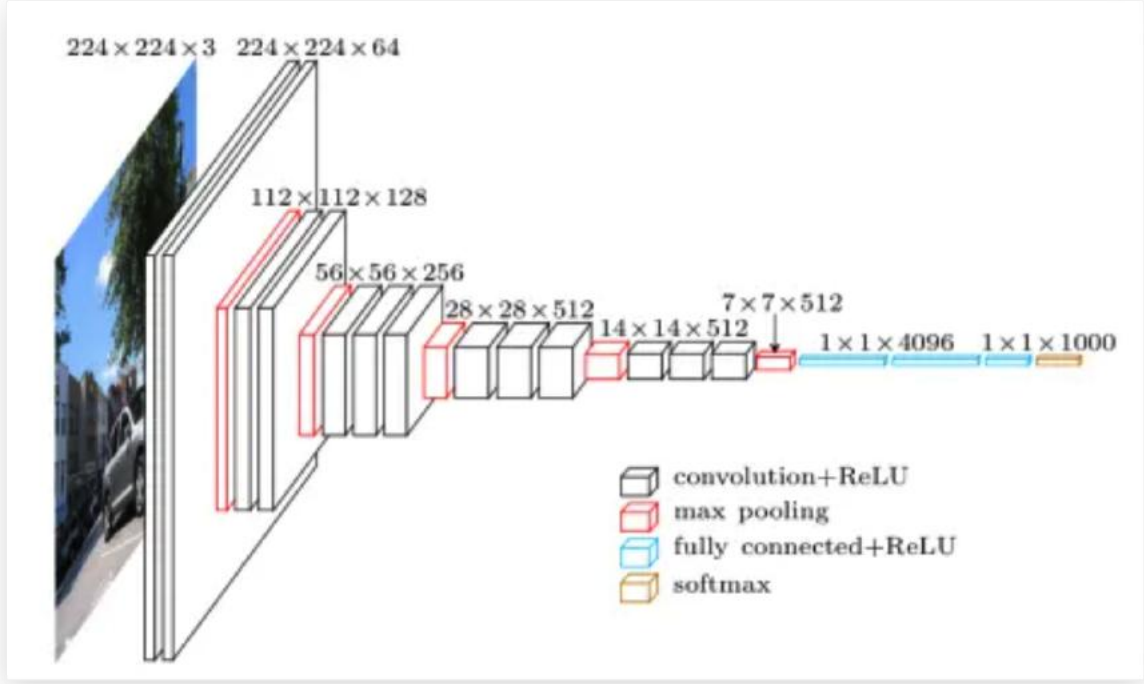


Figure 2.9: Architecture of VGG, adopted from [5].

2.5.3 ResNet

ResNet further extends the number of layers up to 152 with the help of residual blocks [6] and achieved the best top-1 classification performance in ImageNet at that time. Although deep convolutional neural networks can extract levels of features, the training process can be hampered due to the vanishing or exploding gradients. This problem has been largely mitigated by the introduction of residual learning, a learning method that allows consecutive layers to have shortcuts above them to retain identity knowledge from shallower layers as is shown in Figure 2.10. ResNet outperforms previous deep learning models significantly and is therefore applied to radar gait recognition in [10].

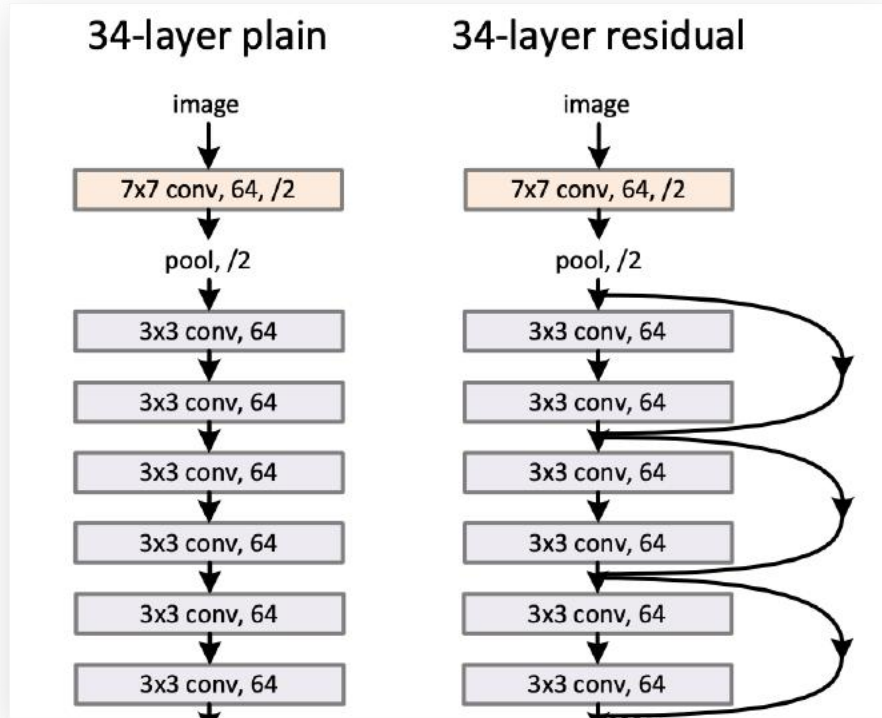


Figure 2.10: Architecture (partial) of ResNet [6]. See full architecture in Appendix A.

2.5.4 ShuffleNet

ShuffleNet was introduced by Zhang *et al.* in 2017 [7]. It is designed for devices with low computational power such as mobile phones and embedded systems. Unlike traditional convolutional neural networks which use standard convolution kernels sliding over the whole image for each channel, the ShuffleNet utilises pointwise group convolution instead of standard convolution and channel shuffle for accelerating the training process. The performance of ShuffleNet is promising in computing speed where it is 13 times faster than AlexNet, while the accuracy is still considerable. The special architecture of ShuffleNet can be viewed in Figure 2.11.

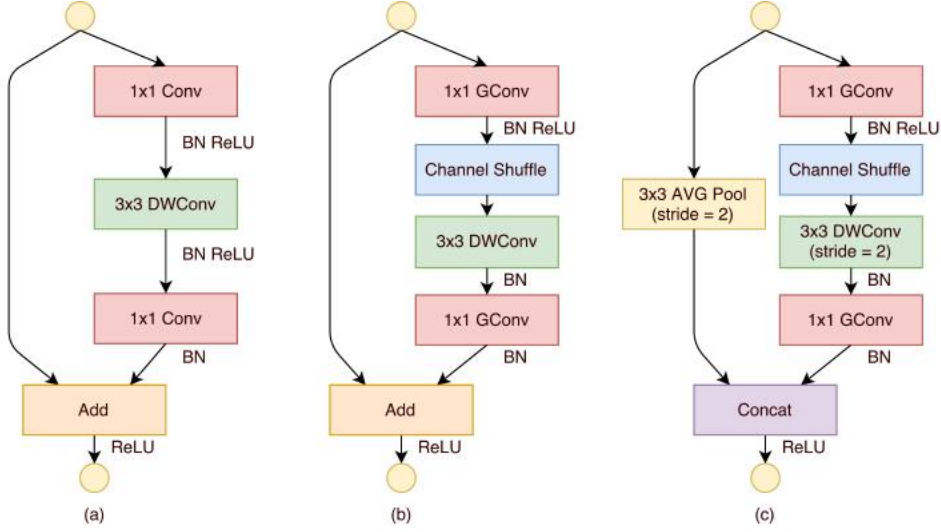


Figure 2.11: Main architecture of ShuffleNet [7].

2.5.5 Vision Transformer

Vision Transformer (ViT) is proposed by Dosovitskiy *et al.* in 2020 and for the first time brings the raw transformer from natural language processing (NLP) to the field of image processing [8]. The Vision Transformer tries to use the attention mechanism in the transformer to understand the meaning of images. This network also treats images as tokens or words in NLP by slicing images into patches of 16×16 to reduce the computational cost induced by too many pixels. Those patches are then sent to a standard transformer for computing their relationship. The learnable class tokens embedded in patches are finally extracted to an MLP for classification. A Vision Transformer model pretrained in a large image dataset can obtain comparable performance although no convolution layers are used. An illustration of Vision Transformer is shown in Figure 3.5.

2.6 Radar-based Gait Recognition

Traditional mDS-based gait recognition systems usually adopt a two-step framework [42–44]. The temporal or statistical time-frequency features are firstly extracted from mDS representations such as spectrograms, and common classifiers are then used to predict the

targets. For instance, the support vector machine (SVM) [45], the k -nearest neighbours (k NNs) classifier [42–44] and the Naïve Bayes (NB) classifier [43, 44, 46] are three popular choices.

Recently, the rapid development of deep learning methods has enabled neural networks to learn features automatically and perform well in radar gait recognition. In [27], authors first used a deep convolutional neural network (DCNN) to perform human detection and activity classification on mDS instead of conventional ways that use hand-crafted features. Similar work has been done by Cao *et al.* [9], who use a DCNN in classifying mDS and achieved an accuracy of 68.9 in a 100-target dataset. To fully exploit the casual relationships of the evolution of gait signal, Addabbo *et al.* [10] proposed a temporal convolutional network (TCN) to recognize people by gait dynamics. The convolution architecture of TCN is characterized by casualness, and the length of the output is constant. A two-level neural attention layer stack also enables the network to pay attention to subsets of characteristics.

Moreover, Bai *et al.* [16] have designed a dual-channel DCNN that recognises human gaits by focusing on a limb and torso aspects independently utilising two sliding window lengths in processing short-time Fourier transformed (STFT) spectrograms. To distinguish individuals from STFT spectrograms, Chen *et al.* [47] have proposed a voted mono-static DCNN and a multi-static DCNN, which combine three pretrained VGG-f networks with dimension reduction layers and fully connected layers, then fuse the results using binary voting or a fusion layer.

Chapter 3

Methodologies

This chapter explains the proposed methodology which solves the radar gait recognition problem in depth. Section 3.1 gives an overview of the novel architecture. Section 3.2 describes how data was processed from raw radar signal to usable mDS representations. In Section 3.3, the complementary nature of spectrogram and CVD is explained. Vision transformer is introduced and explained in Section 3.4. Section 3.5 details the architecture of the proposed attention-based dual-stream Vision Transformer. Attention-based fusion is explained in Section 3.6. Finally, realisation details of radar signal processing and network development are discussed in Section 3.7.

3.1 Overview

The proposed ADS-ViT for radar gait recognition is shown in Fig. 3.1(a). To better exploit the discriminant information embedded in the radar signal, two initial feature representations, spectrogram [19, 20] and CVD [21] are jointly utilised in the proposed framework. Two subnetworks, spectrogram stream and CVD stream, are used to extract features from each of these two representations, respectively. For spectrograms and CVDs, different image patches correspond to different frequency bands. To robustly extract low-level features from these frequency bands, a network based on the Vision Transformer (ViT) [8] is designed, where the local features are encoded through the Position and Patch Embedding mechanism of ViT. ViT could also capture the high-level semantic information across different frequency bands by utilising the self-attention mechanism in the Transformer Encoder of ViT. An attention-based fusion mechanism is proposed to

find the most discriminant features, and fuse the features of two streams into one feature vector. Finally, a fully-connected layer is used to predict the person's identity given the radar signal.

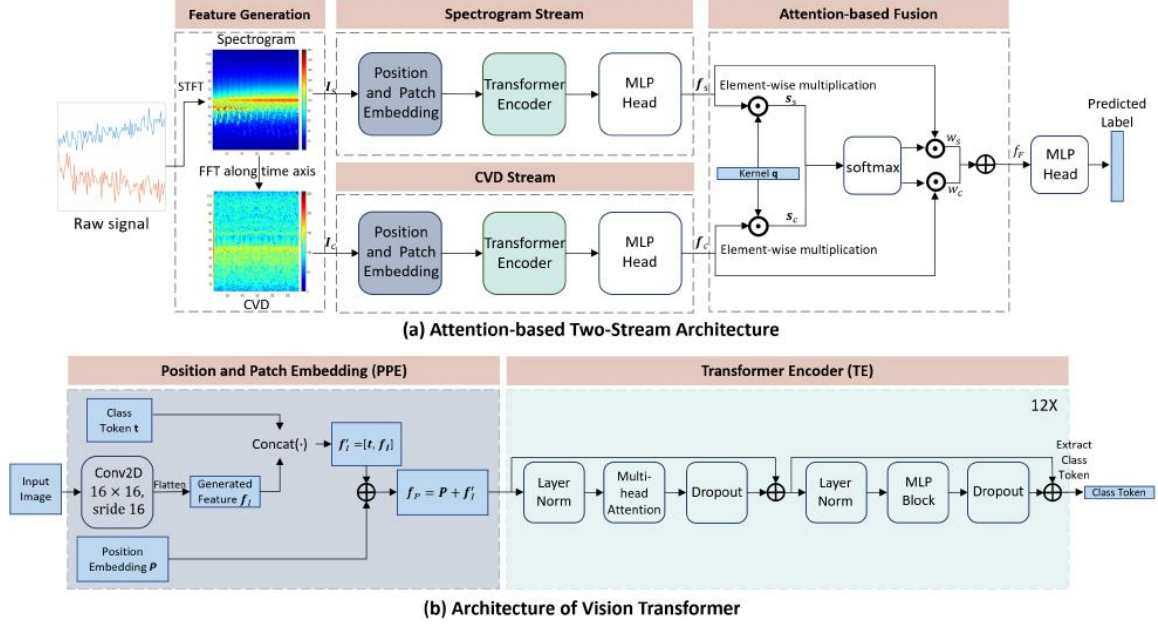


Figure 3.1: (a) shows the overview of the whole architecture; (b) details the pipeline of ViT by illustrating its position and patch embedding and transformer encoder.

More specifically, the proposed ADS-ViT network can be formulated as a quadruplet $\mathcal{Q} = (\mathcal{V}_s, \mathcal{V}_c, \mathcal{F}, \mathcal{C})$, where $\mathcal{V}_s$ and $\mathcal{V}_c$ are the ViT networks for spectrogram and CVD respectively, as shown in Fig. 3.1(b), $\mathcal{F}$ is an attention-based fusion network and $\mathcal{C}$ is the classifier at the end. Denote the generated spectrogram and CVD as I_s and I_c , respectively. The features f_s from the spectrogram stream and f_c from the CVD stream are generated as:

$$f_s = \mathcal{V}_s(I_s), \quad (3.1)$$

$$f_c = \mathcal{V}_c(I_c), \quad (3.2)$$

The attention based fusion layer would then take the two output feature maps and generate a fused feature f_F as:

$$f_F = \mathcal{F}(f_s, f_c). \quad (3.3)$$

Finally, the label $\hat{l}$ is predicted by the classifier $\mathcal{C}$ as:

$$\hat{l} = \mathcal{C}(\mathbf{f}_F). \quad (3.4)$$

3.2 Data Processing

A K-MC1 radar transceiver and an ST200 evaluation system launched by RFbeam Microwave GmbH were employed to collect a radar gait dataset by the authors. The raw radar signal is originally recorded in the form of complex data or in-phase/quadrature (IQ) data which enables a degree of freedom to recover phase.

The signal $s(t)$ is then segmented into M overlapping frames $\{\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{M-1}\}$, where each frame $\mathbf{x}_i = \{x_i[n], n = 0, 1, \dots, N-1\}$ is a column vector of length N . These M frames form a synthetic image $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_{M-1}]$ of size $M \times N$. Then, the discrete Fourier transform $\mathbf{f}_i = [f_{i,0}, f_{i,1}, \dots, f_{i,N-1}]$ of $\mathbf{x}_i$, $\mathbf{f}_i = \mathcal{F}\{\mathbf{x}_i\}$ is computed as:

$$f_{i,k} = \sum_{n=0}^{N-1} x_i[n] \exp \left\{ -j2\pi \frac{kn}{N} \right\}, \quad (3.5)$$

where $\mathcal{F}\{\cdot\}$ denotes the Fourier transform. The magnitude $\mathbf{S} = \{|\mathbf{f}_{i,k}|^2, i = 0, \dots, M-1, k = 0, \dots, N-1\}$ forms the spectrogram of size $M \times N$. An illustration of spectrogram used as input to our network can be seen in Figure 3.2.

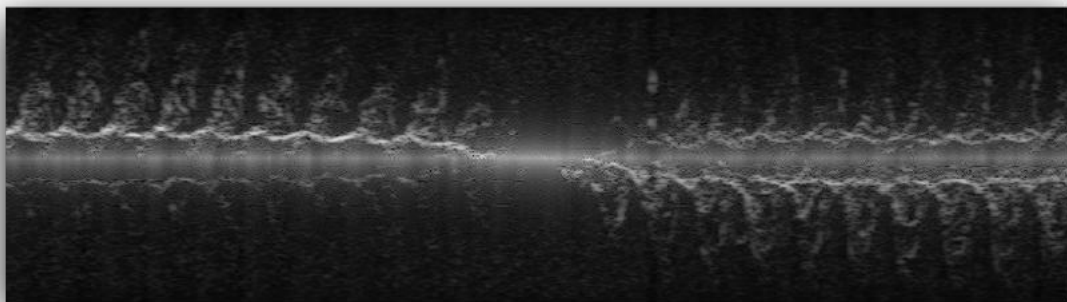


Figure 3.2: The spectrogram shows a person walking towards the radar and away from it, forming a somehow centrosymmetric image.

The spectrogram demonstrates the time-varying feature of gait cycles but could be too

long for a standard neural network to learn. Our approach is to slice the original spectrogram into multiple frames of 115 data points, with a stride of 10 to allow overlapping. A sample sliced spectrogram segment is shown in Figure 3.3.

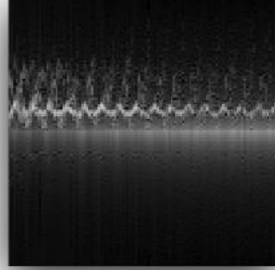


Figure 3.3: A segment sliced from a complete spectrogram.

The CVD is obtained by taking the Fourier transform on the spectrogram along the time dimension for each Doppler frequency bin, i.e.

$$D = F_t\{S\}, \quad (3.6)$$

where D stands for CVD and $F_t\{*\}$ means the Fourier transform along the time dimension. An illustration of CVD used as input to our network can be seen in Figure 3.4.

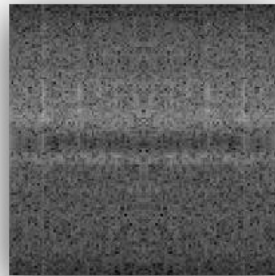


Figure 3.4: The CVD shows the frequency of certain speeds of body parts.

3.3 Complementarity of Radar Signal Representations

Spectrogram: Tahmoush and Silvius [28] modelled Doppler of each body part of a walking human as sinusoidal modulation in the spectrogram. The model reveals the velocities

of each body part which is instrumental in determining the necessary human gait characteristics. The fundamental characteristics are the mean Doppler velocity and the size of the torso variation in the micro-Doppler. It has been found that the mDS of a person is relatively consistent and different walking people lead to clearly discriminative mDS [48]. Time-frequency spectrogram can well manifest such time-varying human gait characteristics. A sample spectrogram is shown in Fig. 2.5.

Cadence velocity diagram (CVD): The Cadence velocity diagram has been proposed by [21] to analyse micro-Doppler signatures. After deriving the spectrogram, the CVD is obtained by taking the Fourier transform on the spectrogram along the time dimension for each Doppler frequency bin.

The derived CVD, as shown in Figure 2.7, is a matrix with rows containing Doppler frequency (or targets' speed) and columns containing cadence frequency, which measures how frequently different frequencies appear in the signal over time [29]. In other words, it illustrates the useful information of the repetition of velocities, which provides complementary knowledge to the spectrogram and can be another key identification feature of a walking person.

3.4 Backbone Vision Transformer

Compared to commonly used convolutional neural networks such as VGG [5], AlexNet [4], MobileNet [49] and ResNet [6], Vision Transformer (ViT) [8] can reshape the spectrogram into several local patches and use attention mechanism to obtain global knowledge of patches. Unlike scenarios of real-world images where an object could be positioned anywhere in an image but mean the same, patches of spectrogram and CVD converted from radar signal have their unique physical meanings when positioned in different locations in a diagram. This is because each patch contains knowledge about its specific area according to its time and frequency dimension for spectrogram or frequency dimensions for CVD. Even patches similar in representation could be totally different physically based

on their frequency levels and temporal position. Instead of convolutional kernels in CNN, patches are therefore ideally used to reserve the positional information of signal representations as well as the relationship between them. In our method, ViT is responsible for splitting diagrams in patches and embedding their positions, and the multi-head attention mechanism in the transformer encoder could further extract the reserved physical information, processing physical meanings in the same way as natural language.

The architecture of ViT is depicted in Figure 3.5. The input image is first split into 16×16 separating patches, and each patch is then flattened to a 768-length vector. Then classification tokens are concatenated and position embedding is conducted by adding a position matrix, resulting in an overall 197×768 matrix. The sequence of vectors is then fed to a standard transformer encoder, and this would be executed 12 times for the ViT base model. The transformer encoder module contains a multi-head self-attention layer, which is global and can process the relationship between patches. The output feature vector of the transformer encoder after classification token extraction is in the size of 1×768 . An MLP Head containing a fully-connected layer is responsible for taking the above feature vector and classification.

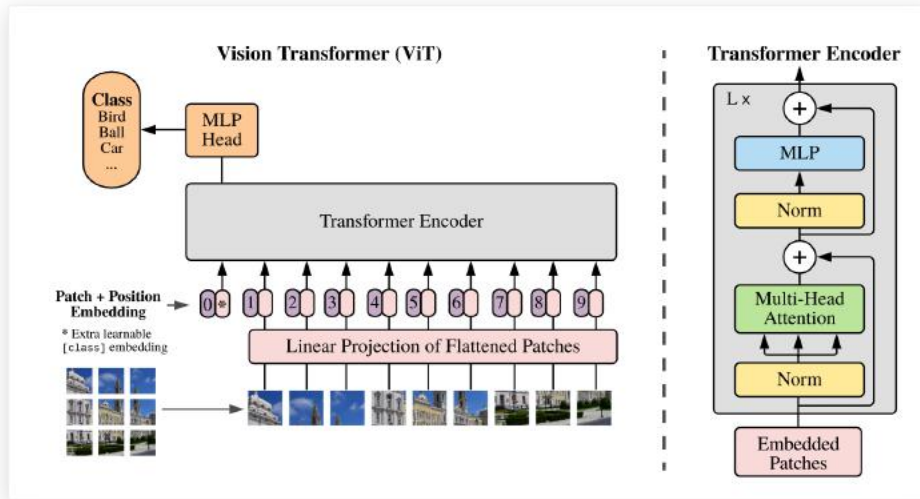


Figure 3.5: Architecture of Vision Transformer [8].

3.5 Dual-stream Vision Transformer

Two network branches based on the Vision Transformer are designed to extract features from spectrograms and CVDs, respectively. The ViT network consists of two parts: Position and Patch Embedding (PPE) and Transformer Encoder (TE), as shown in Fig. 3.1(b). The original size of spectrogram/CVD is 115×115 . To fit the pretrained ViT network, the input spectrogram/CVD is converted to a 3-channel image $\mathbf{I} \in \mathcal{R}^{224 \times 224 \times 3}$, by replicating the resized grayscale image for three channels. In the PPE stage, $\mathbf{I}$ is convolved into the feature map of size $14 \times 14 \times 768$ via a 16×16 kernel with a stride of 16, and flattened to the features $\mathbf{f}_I \in \mathcal{R}^{196 \times 768}$. Similarly as in [8], the learnable classification token $\mathbf{t} \in \mathcal{R}^{1 \times 768}$ is prepended to $\mathbf{f}_I$ as:

$$\mathbf{f}'_I = [\mathbf{t}, \mathbf{f}_I]. \quad (3.7)$$

The position information is embedded by adding a learnable position matrix $\mathbf{P} \in \mathcal{R}^{197 \times 768}$ for retaining positional knowledge of each patch, similarly as in [8], which results in the feature map $\mathbf{f}_P \in \mathcal{R}^{197 \times 768}$ with position embedding to the Transformer Encoder as:

$$\mathbf{f}_P = \mathbf{P} + \mathbf{f}'_I. \quad (3.8)$$

The TE module consists of 12 repeating blocks, where each contains a multi-head self-attention layer. This module could learn the global relations between patches. The learned knowledge is represented in the classification tokens for all patches and used as the output of the ViT network.

Unlike real-world images in which an object could be positioned anywhere in an image but interpreted the same, patches of spectrogram or CVD generated from the radar signal have unique physical meanings when positioned at different locations in the image. Each patch in spectrogram or CVD contains information in different frequency bands at different time instances (or different cadence velocities for CVD). Even patches similar in appearance could be physically different based on their frequency bands and temporal positions.

The commonly-used convolutional neural networks such as AlexNet [9], VGG [10] and ResNet [10] often use the same convolutional kernel across the whole image, while the proposed ADS-ViT treats patches differently, *i.e.*, it splits the spectrogram/CVD into patches and embeds their positions so that the information embedded in different frequency bands could be properly aligned and effectively extracted. This partially justifies the superior performance of the proposed ADS-ViT over other models. The multi-head self-attention mechanism in the Transformer Encoder could further extract the reserved discriminant information from each patch and process this information globally to encode the global relations among patches.

3.6 Attention-based Fusion

Inspired by Chen *et al.* [50], an attention-based fusion architecture shown in Fig. 3.1(a) is developed to fuse the complementary features extracted from both spectrogram and CVD. The target of the attention-based feature-level fusion is to find a set of weights $\{\mathbf{w}_i \in \mathcal{R}^{1 \times 768}\}$ for the features $\{\mathbf{f}_i \in \mathcal{R}^{1 \times 768}\}$ to obtain an aggregated feature $\mathbf{f}_a \in \mathcal{R}^{1 \times 768}$:

$$\mathbf{f}_a = \sum_{i=1}^n \mathbf{w}_i \odot \mathbf{f}_i, \quad (3.9)$$

where $\odot$ denotes element-wise multiplication, n is the number of features and $n = 2$ in this paper. In the attention-based fusion model, the kernel $\mathbf{q} \in \mathcal{R}^{1 \times 768}$ is required to be trained. The process begins with the element-wise multiplication between the feature vector $\mathbf{f}_i$ and the kernel $\mathbf{q}$:

$$\mathbf{s}_i = \mathbf{q} \odot \mathbf{f}_i, \quad (3.10)$$

where $\mathbf{s}_i \in \mathcal{R}^{1 \times 768}$ are the confidence scores for feature representations. A softmax function is then applied to $\mathbf{s}_i$ to assure that the derived weights $\sum_i \mathbf{w}_i = \mathbf{1} \in \mathcal{R}^{1 \times 768}$:

$$\mathbf{w}_i = e^{\mathbf{s}_i} \oslash \sum_j e^{\mathbf{s}_j}, \quad (3.11)$$

where $\oslash$ denotes element-wise division. The fused feature vector is finally generated using Eq. (3.9). This attention-based fusion could well highlight the most discriminant features by training the kernel function $\mathbf{q}$.

3.7 Realisation

3.7.1 Radar Signal Processing

The original raw radar signal was collected as in-phase/quadrature (IQ) data and stored using MATLAB. For the convenience of data processing, we also use MATLAB as the major tool in signal processing. As people walk towards or away from the radar, the magnitude of the received radar signal would be recorded by the sensor with a certain frequency. Additionally, the magnitude of the radar signal consists of two channels and is thus presented in a complex number form. The first step is to separate the two channels for further transformation. The separated radar signal is illustrated in Figure 3.6, where the horizontal axis is time and the vertical is magnitude. The following describes the implementation of radar signal processing.

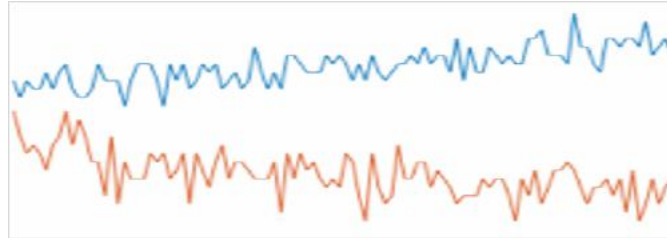


Figure 3.6: Illustration of separated raw signal.

1. The first step of processing the signal is to extract the two channels and decimate the complex data to reduce the data size.
2. Next, certain indexes for useful signals are assured for windowing and taking fast Fourier Transform. Augmentation methods are applied to obtain more discriminative data.
3. Finally, standard normalisation is applied, and spectrogram is obtained.

The spectrogram obtained above contains more than 30 seconds of gait data, while only segmented gait frames within 2 seconds are needed. The spectrogram is then segmented into several frames with overlap and saved in PNG format. CVD is obtained almost at the same time after segmentation but processed by another fast Fourier Transform. The CVD obtained is saved using the same order as the spectrogram.

For the convenience of training neural networks using transfer learning, the obtained spectrogram and CVD are normalised and converted to RGB images by repeating for the three channels. The sample segmented spectrogram and CVD represented in the heat map are shown in Figure 3.7.

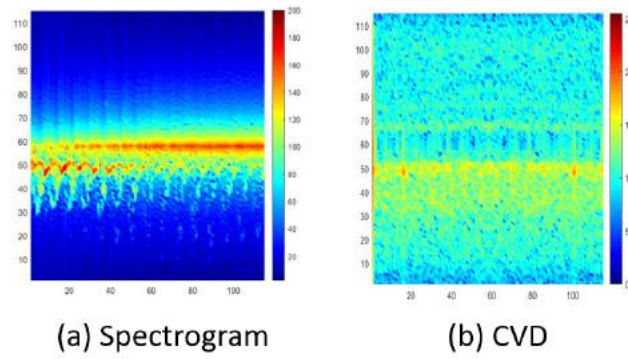


Figure 3.7: Processed spectrogram and CVD.

3.7.2 Neural Network Implementation

The standard implementation from PyTorch of AlexNet, VGG and ResNet are utilised as state-of-the-art methods for comparison. However, the dual-stream architecture is designed and implemented by authors using PyTorch.

This architecture has two streams of the same network whose classification layers are deleted and feature extraction layers are kept. Two feature outputs from both streams then are fed into an attention layer for fusion, keeping the most useful features from the two streams. The fused feature remains the same size and is then fed into the last classification layer for final recognition.

Training a dual-stream neural network means a large amount of backpropagation, and since ViT does not utilise convolutional kernels for reducing parameters, this would be a fairly slow process. However, one advantage of this architecture is that the pretrained feature of those backbone networks can be easily transferred to this architecture and largely accelerate the training process. Furthermore, the architecture also supports taking extracted features as input, *i.e.*, backbone networks can be pretrained for feature extraction and save features extracted directly as the input of the dual-stream network. Without complex computational operations, this considerably reduces the training time and provides high flexibility as two backbones can be adjusted respectively. Additionally, the framework authors designed is compatible with not only ViT but also AlexNet, VGG and ResNet for necessary comparison. Two feature outputs from both streams then are fed into an attention layer for fusion, keeping the most useful features from the two streams. The fused feature remains the same size and is then fed into the last classification layer for final recognition.

3.7.3 Network Training Implementation

Training a neural network can be hard as hyperparameters are required to be tuned carefully for each network architecture. This long process also needs a well-designed network training architecture to help.

We utilised two GPUs to calculate in parallel to accelerate the training. *DataParallel* function could distribute data almost equally to two GPUs for tensor calculation. Numbers of experiments are done for finding better solutions, and this process also requires recording relevant parameters and results. TensorBoard from TensorFlow is utilised as the visualisation and recording tool for parameters and metrics such as accuracy and loss.

Figure 3.8 shows an example of the visualisation using TensorBoard. The x-axis stands for epoch and will record learning rate, validation accuracy and loss in real-time. This largely helps authors to monitor the real-time status of training and can help compare

the performances of different models intuitively.

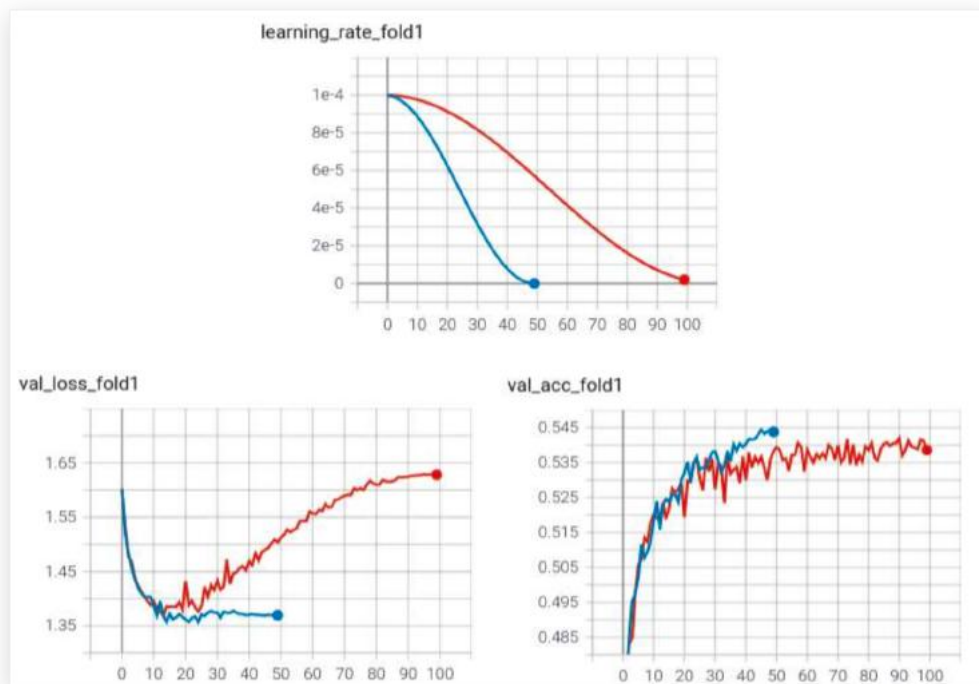


Figure 3.8: Two runs illustrated using TensorBoard with their learning rate change, validation accuracy and loss.

Chapter 4

Experimental Results

This chapter first gives an overview of the radar gait dataset and then outlines and explains the experimental results of both the ablation study and comparisons to state-of-the-art methods. Section 4.1 introduces the collection and basic information of the dataset. Section 4.2 lists main experimental settings authors adopted. Section 4.3 compares the proposed approach with other state-of-the-art radar gait recognition methods and shows the effectiveness of our approach. Section 4.4 illustrates the performance of using ViT on spectrogram and CVD and the proposed attention-based dual-stream architecture with the other three backbones, AlexNet, VGG and ResNet. Finally, the confusion matrix of prediction by ADS-ViT is analysed in Section 4.5 to summarise the performance of the proposed method.

4.1 Dataset Overview

There is no publicly available dataset for radar gait recognition, and hence a K-MC1 radar transceiver and an ST200 evaluation system launched by RFbeam Microwave GmbH are employed to collect a radar gait dataset by the authors. There are two data collection sessions at least two weeks apart. In each session, each volunteer walks 10 sequences. (Some only participate in the first session.) The dataset consists of 1670 walking sequences from 98 volunteers. For each volunteer, 50% of sequences are randomly selected as the training set and the rest are used for testing. In each sequence, a volunteer walks away from the radar along a corridor about 40 meters long, turns around and walks back towards the radar, lasting about 30 seconds. The sampling rate of the original signal is

125k. After a decimation of 64, the sampling rate is about 1.95k. The spectrogram is built using 128 sample points with an overlapping ratio of 90%. After removing the cluster and non-informative high-frequency components, the spectrum has 115 data points. To enrich the dataset, each sequence is cut into multiple frames of 115 data points, with a stride of 10. As a result, a total number of 45,768 frames of size 115×115 are generated, in which 22,894 are used for training and 22,874 for testing. A glance of the spectrogram and CVD frames in our dataset is shown in Figure 4.1 and 4.2.

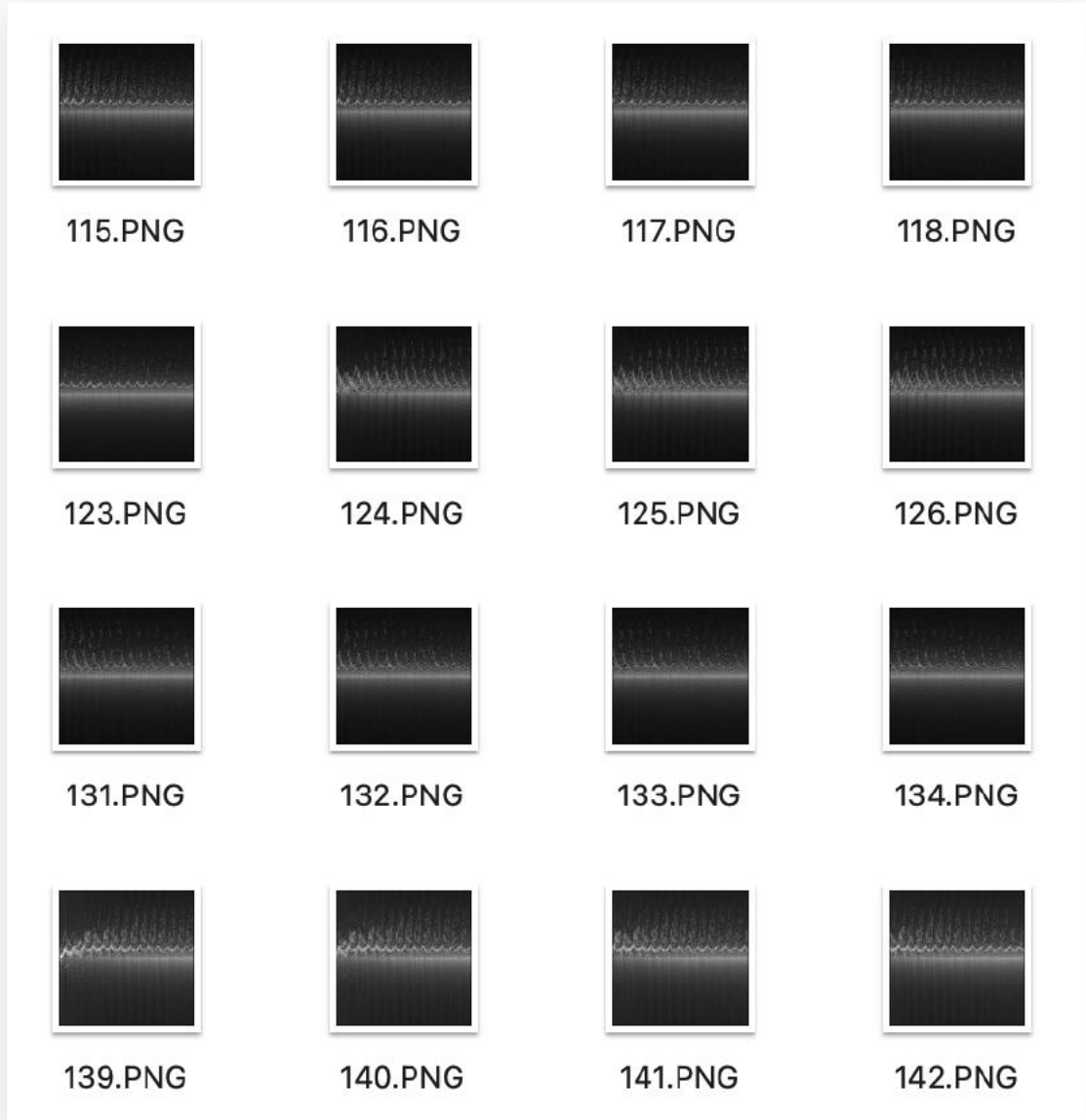


Figure 4.1: A glance of spectrogram dataset.

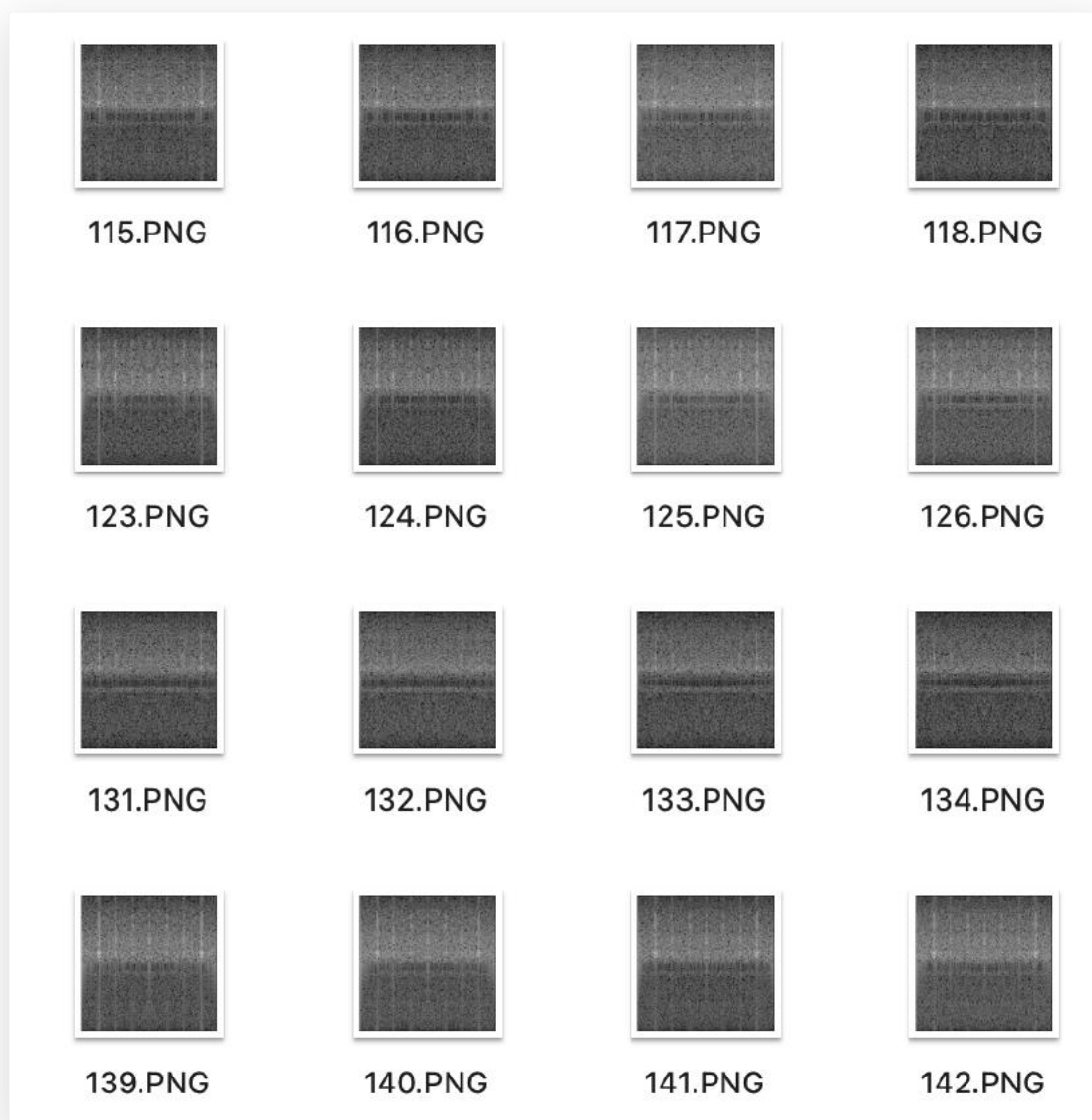


Figure 4.2: A glance of CVD dataset.

4.2 Experimental Settings

AlexNet has been used as the backbone of the Deep Convolutional Neural Network on spectrograms of the radar signal to identify people [9]. In [10], VGG16 [5] and ResNet18 [6] have been used for radar gait recognition. These three approaches are state-of-the-art methods for radar target recognition. They are implemented and evaluated on our benchmark dataset. As the proposed ADS-ViT utilizes both spectrogram and CVD, AlexNet [9], VGG16 [10] and ResNet18 [10] are applied on the CVD for comparison as well. The Vision Transformer we adopted is ViT-Base with model specifications listed in Table 4.1.

Table 4.1: Variants details of Vision Transformer Base

Model	Layers	Hidden size	MLP size	Heads
ViT-Base	12	768	3072	12

The size of frames is 115 initially and resized to 224 to fit the networks. Stochastic Gradient Descent is used as the optimizer of all models with a momentum of 0.9 and a weight decay of 5×10^{-5} . The cross-entropy loss function is used. For learning parameters, a linear learning rate warmup and decay strategy are used with a learning rate scheduler, and the initial learning rate is set to 0.01 for ResNet, 0.001 for AlexNet and 0.0001 for VGG following the parameter settings in [10] and [9] for a fair comparison. The batch size is 128 for all models. The proposed ADS-ViT initialize ViT networks with pretrained weights for both spectrogram and CVD streams. Models are trained for 500 epochs to guarantee convergence, and the optimal performance on the test set is reported. The network training-related settings are listed in Table 4.2.

Table 4.2: Training-related Settings

Network	AlexNet	VGG	ResNet	ViT
Learning Rate	0.001	0.0001	0.01	0.01
Learning Rate Scheduler	Cosine Annealing			
Optimiser	Stochastic Gradient Descent with Momentum 0.9			
Batch Size	128			
Epoch	500			

The experimental environment we adopted is listed in Table 4.3.

Table 4.3: Experimental Environment (See full environment setting in Appendix B)

OS	Ubuntu 18.04.5
CPU	Intel(R) Xeon(R) Silver 4108 CPU @ 1.8GHz $\times 32$
GPU	GeForce RTX 2080 Ti (12GB) $\times 2$
Memory	128GB
NVIDIA Driver	460.91.03
Python	3.8.8
PyTorch	1.8.0

4.3 Comparison to the State-of-the-art Approaches

The comparisons to state-of-the-art models are summarised in Table 4.5. Most existing models are applied on spectrograms, *e.g.*, AlexNet [9], VGG16 [10] and ResNet18 [10]. As the proposed method utilises both spectrogram and CVD, these three CNN models are applied on the CVD as well.

Table 4.5 shows that the proposed ADS-ViT significantly outperforms all compared methods. As for methods applying on the spectrogram, AlexNet and VGG perform similarly with an accuracy of about 70%, while ResNet achieves a much better performance of 85.56% with the help of its unique residual blocks. However, when applying on CVD, AlexNet and VGG perform slightly better than on spectrogram, while the accuracy of ResNet drops by about 5% to 80.73%. The best accuracy so far is achieved by ResNet on the spectrogram with 85.56%, on the other hand, the proposed ADS-ViT further improves the classification accuracy to 91.02%. The achieved significant performance gain is attributed to the proposed ADS-ViT in two folds. First, by dividing the spectrogram/CVD into patches, the proposed model extracts the local features from different frequency bands using the Position and Patch Embedding mechanism. Secondly, the proposed model extracts the global features across frequency bands using the self-attention mechanism embedded in the transformer encoder.

Table 4.4: Comparisons to state-of-the-art AlexNet [9], VGG16 [10] and ResNet18 [10]. These models are implemented and evaluated on both spectrograms and CVDs on our benchmark dataset for comparison.

Method	Accuracy
DCNN-AlexNet on Spectrogram [9]	71.56%
VGG16 on Spectrogram [10]	69.24%
ResNet18 on Spectrogram [10]	85.56%
DCNN-AlexNet on CVD	72.44%
VGG16 on CVD	79.83%
ResNet18 on CVD	80.73%
Proposed ADS-ViT	91.02%

4.4 Ablation Study

The Vision Transformer itself can work as a fairly effective feature extractor and classifier on radar gait recognition tasks because of its special nature of treating images as patches. The experiments conducted show that ViT on spectrogram can achieve an accuracy of 90.03%, and ViT on CVD can obtain a top-one accuracy of 80.14%. The result of ViT on spectrogram outperforms all the other state-of-the-art methods on both representations. This is due to that the three state-of-the-art methods all use convolution layers for extracting features, which ignores the physical meanings embedded in radar representations. ViT, however, does not have this drawback and can still obtain enough features for better classification.

Table 4.5: Results of ViT [8] on spectrogram and CVD and ResNet are shown in the table. Experiments of AlexNet, VGG, ResNet and ViT using the proposed attention-based dual-stream architecture are also conducted and demonstrated in the table.

Method	Accuracy
ViT-base 16x16 on Spectrogram	90.03%
ViT-base 16x16 on CVD	80.14%
Attention-based Dual-stream AlexNet	82.86%
Attention-based Dual-stream VGG	86.57%
Attention-based Dual-stream ResNet18	87.08%
Proposed ADS-ViT	91.02%

The proposed attention-based dual-stream architecture can be adopted with various backbone networks including AlexNet, VGG and ResNet. The results of experiments also show that the classification performance can be significantly improved by the proposed attention-based dual-stream architecture. Especially for AlexNet, the accuracy using attention-based dual-stream architecture exceeds the original method using spectrogram by about 10%. It is also considered effective for VGG and ResNet, which have 6.74% and 1.52% increment in accuracy respectively. This considerable improvement can be attributed to the efficient fusion ability of attention-based aggregation. To conclude, the results of experiments on ViT and attention-based dual-stream architecture show their effectiveness respectively, while our proposed ADS-ViT outperforms all the other methods and achieves an accuracy of 91.02%.

4.5 Confusion Analysis

High accuracy may not be able to fully convince the good performance of the proposed method. There are 98 classes in our dataset contributed by 98 volunteers, and it is possible that some of the classes are not predicted well while the correctly classified targets can still be large, contributing to generally good accuracy. To further study the actual performance of the proposed method, we also use another metric confusion matrix for evaluation.

The confusion matrix of prediction by ADS-ViT is shown in Figure 4.3, where deeper colour means fewer samples and vice versa. It summarises prediction results in a 98×98 matrix. The number of correct and incorrect predictions is displayed in the matrix and broken down by classes. The diagonal records the number or ratio of how many samples of a class are classified correctly, and the off-diagonal area represents samples that confuse the classifier. The values in the confusion matrix are normalised by dividing the total number of samples. Compared to the off-diagonal area, the diagonal is filled with light colours, representing most of the samples are classified correctly. The deep black off-diagonal area also indicates that no obvious confusion exists for the model. There are

some deep purple squares on the diagonal, such as classes 2,3 and 12. However, their values are still relatively larger than the values from the same rows and columns. The deeper colour of these classes is due to the small sample size of these classes, so the values in the confusion matrix are relatively deducted because of normalisation. In conclusion, the confusion matrix shows that our proposed method can clearly recognise all 98 classes from the dataset with only small confusion while achieving convincing overall accuracy.

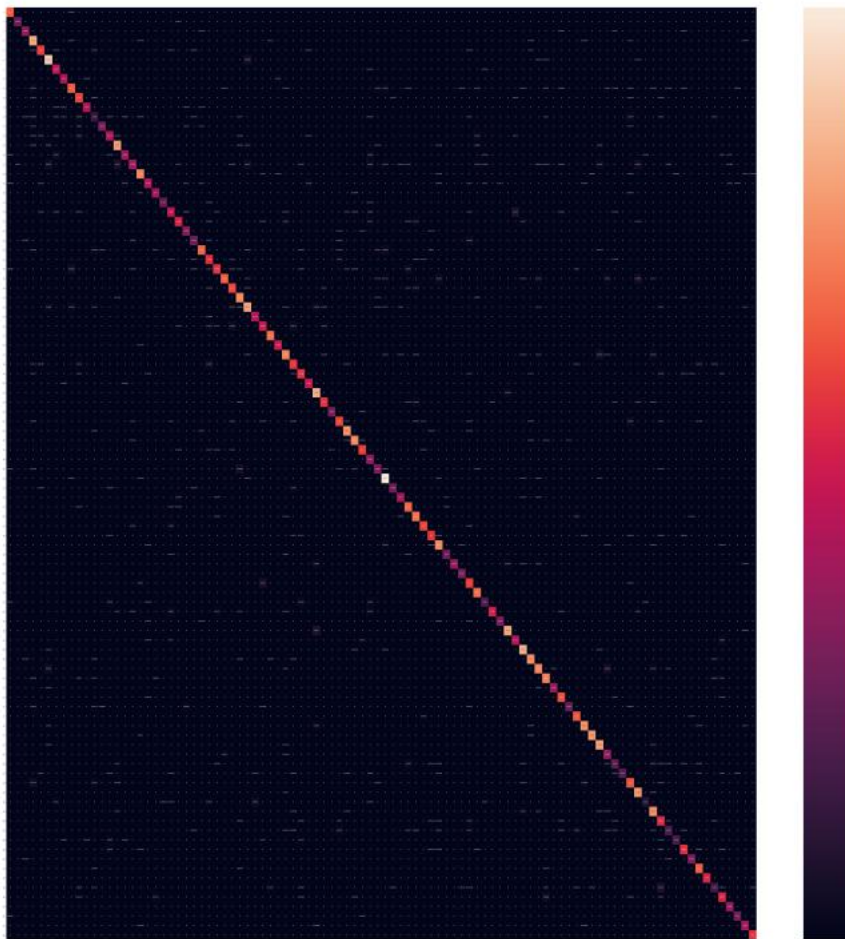


Figure 4.3: Confusion matrix of prediction by ADS-ViT.

Chapter 5

Summary and Reflections

This chapter summarises the whole project and shows relevant reflections. Section 5.1 introduces the strategy and practice of managing the project. Several stages of the project are outlined, and the details of each stage are explained. Section 5.2 contains the reflections on the time management, risk management, failure management of the project as well as experience in conducting formal research. Section 5.3 concludes the achievements of this project and discusses the drawbacks of the system and possible future work.

5.1 Project Management

The timeline of the project is illustrated in a Gantt chart shown below. The chart could automatically present the bar of a corresponding length by entering the start date and duration. The bars in different colours denote different statuses. Purple means complete status, and yellow refers to beyond the plan. Beginning from July 11 to April 4, there are 28 weeks in total, and the plan contains six stages total. The project is deliberately mostly arranged in the summer vacation and first semester for submitting a conference paper.

Until the end of the first semester, the project was overall managed on schedule. The project started from the beginning of summer vacation and all the necessary preparations were made in the first week, including making the project plan and project proposal. As is shown in the Figure 5.2, a thorough literature review was planned to last for one month. This is because the project is related to radar signal and deep learning, which are two unfamiliar areas for me. During the first week of the literature review period,

the fundamental of radar signal and its representations were studied, and an additional week is for learning how to process the raw radar signals. Deep learning is also a new field for me. Understanding the idea of various neural network components such as multi-layer perceptron and convolution layers is the most challenging one for a newcomer. In addition to understanding the concepts of deep learning, the Python implementation was also difficult as it was a new language for me and a number of Python modules were involved, which took me much time to get familiar with. The literature review for deep learning therefore took one more week than planned. The study on gait recognition in literature was postponed as well. During the study of gait recognition, the idea of utilising two representations to improve gait recognition was raised.

As for the design and experiment phase, two representations of the Doppler radar signal were deeply studied. In this phase, the theoretical proof of using the two representations as complementary features was studied to ensure the raised idea in the previous phase was valid theoretically. The study of deep learning in this stage focused more specifically on the methods that used in related works, which took one week. The design and implementation of the network was another challenge, which took around five weeks to implement a fundamental architecture. This is due to the speciality of the idea that two features are processed simultaneously, where the network architecture should be designed significantly different from the common ones. Experiments were conducted immediately after the network was implemented, and evaluation was taken during the experiments as well.

The next stage was for preparing the conference. Different from what was planned, the conference preparation period consisted of iterations of new experiments and verification. The idea was more formally verified by additional experiments and theoretical support. The writing task itself was also challenging as it was my first time engaging in a preparation for a formal conference. The paper format, writing style and many details were learned during this stage, as well as the knowledge about popular computer science conferences. With the help of senior PhD students and the supervisor, the paper was completed

by the beginning of October and accepted by the conference.

An extended idea of adding prior knowledge to the architecture was raised after the submission of the conference paper. The next stage focuses on studying a new asymmetric attention architecture with prior knowledge. This stage contains modifications to the proposed network and more experiments conducted in order to verify the idea.

1. Preliminary work
2. Literature review
3. Design & experiment
4. Conference paper
5. Interim report
6. Final report

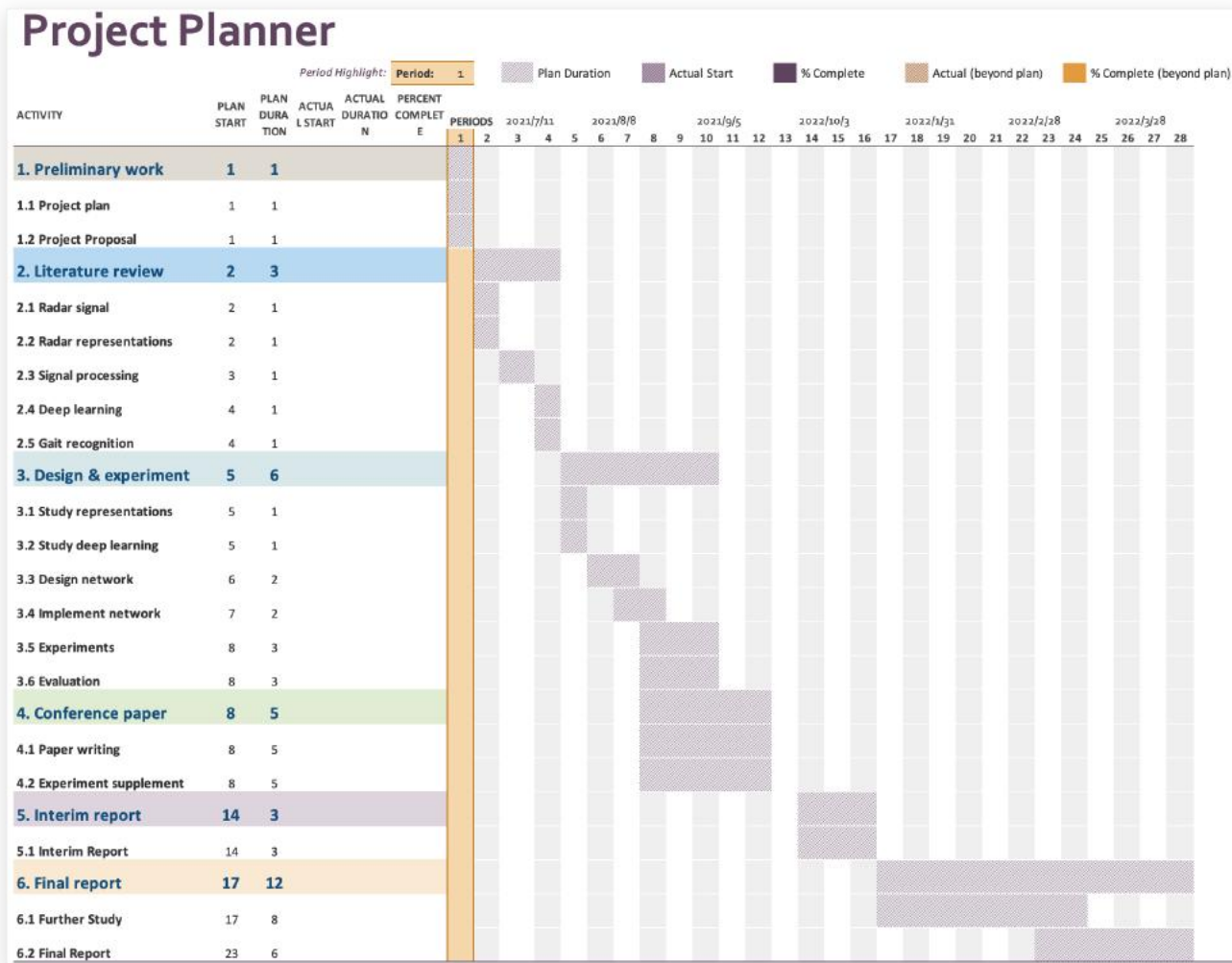


Figure 5.1: Initial Project plan.

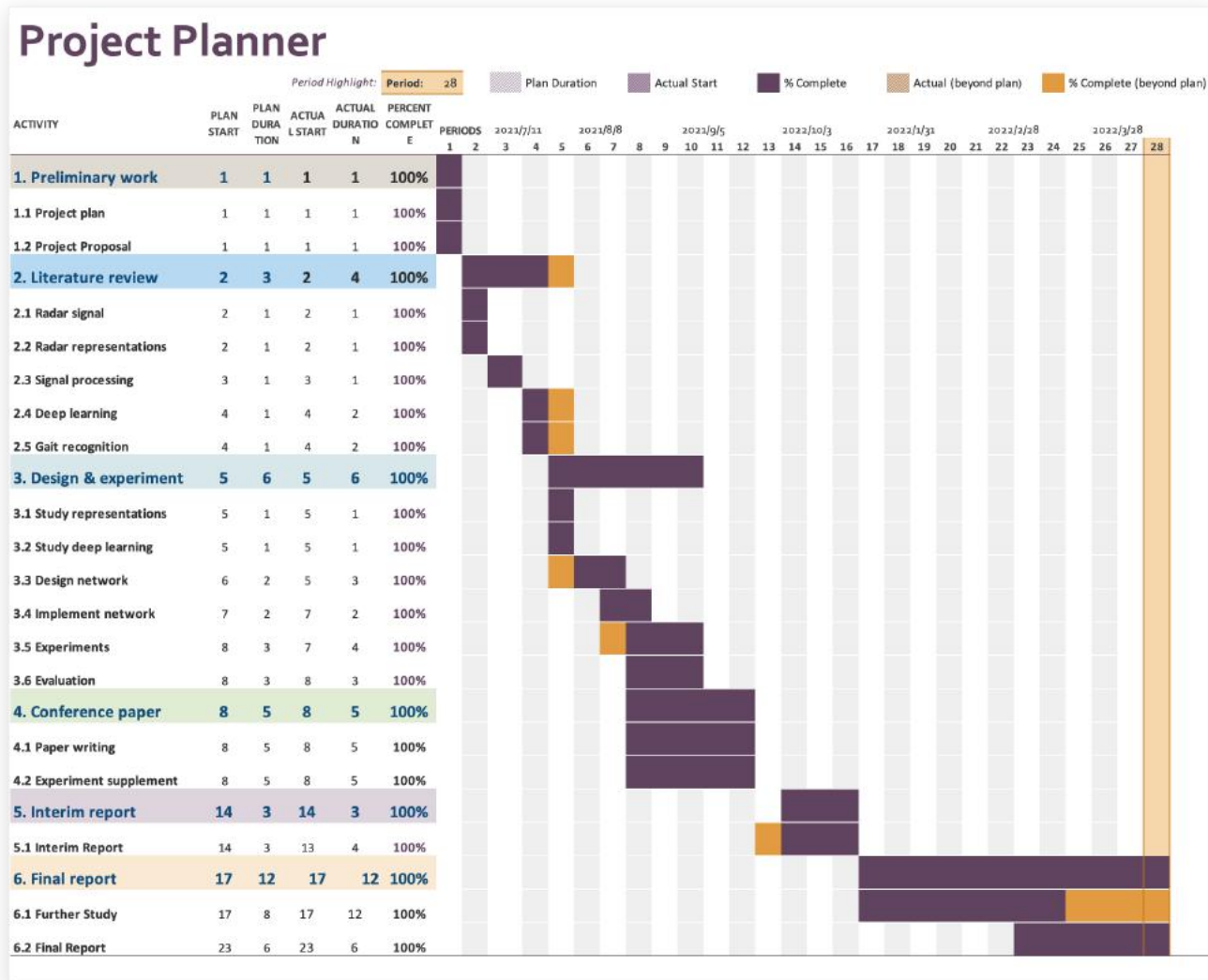


Figure 5.2: Final Project plan.

5.2 Reflections

This section reflects on the project from three aspects. Section 5.2.1 and 5.2.2 discusses what experience I have transferred from other projects in time management and risk management, as well as new problems I met during the project and how I deal with them. Section 5.2.3 describes the lessons I learned from conducting academic research.

5.2.1 Time Management

Positive Experience

Thanks to previous research that I have participated in, I could obtain an overview of research procedures before this project. However, this project is much more formal and needs professional skills. Under the pressure of submitting a conference paper and the time limitation, I was required to absorb knowledge from unknown areas fast and manage the whole project in a smooth way. All of these were challenging my ability to absorb knowledge and management of time.

The goal of publishing our work at a conference dictated that the timing of this project should be carefully scheduled. Since this is my first time working on a conference paper, the potential risks also require me to make sure having enough time for potential adjustments. The task was intense, with only three months between the start of the summer and the end of the submission period, and a series of other commitments such as language exams, front-end projects, and preparation for graduate school while conducting research over the summer. Therefore, organising the summertime can be complicated and challenging. This project taught me how to organise my time, even when time is already tight. A clear goal and direction is always the major thing to keep in mind for managing time. Gantt chart is a good time and goal scheduling tool, which allows me to have a general idea of the overall timeline and goals, and be aware of key time points. This allowed me to be comfortable with the new tasks that were discussed during the meeting

and left enough time to finish writing the conference paper.

In addition to the general organisation of time, I have also done well in managing the time of sub-projects. The time needed for the experimental phase is the most unpredictable, so it requires more careful planning. It is important not only to have the design of the experiment in mind but also to write down the ideas and plans so that parallel work can be organised better. For example, when waiting for the result of the validity of an experimental protocol, it is more efficient to consider the next protocol and start preparing a new database. It is a good idea to define a deadline for testing an idea, and turn to other tasks when time is up to avoid wasting time. This experience has helped me to keep my experiments on track. There are also some unexpected events in the project, such as the sudden need to prepare a presentation and poster for a conference. This is where the margin left in front of me comes in handy and allows me to deal with the unexpected.

Negative Experience

During the first few weeks of the project, I did not realise how heavy the workload would be. I was not active in proceeding with the process and did not make any clear plan. However, several meetings with the supervisor alarmed me to be disciplined and be responsible for the whole project. It is fairly important to be active in the beginning and take responsibility for the project and the group of people I work with.

5.2.2 Risk Management

Positive Experience

Previous experience in the Software Engineering Group Project provides me with precious ideas for managing risks. Various risks exist over the whole period of the project. One of the biggest risks is that the experimental results do not meet theoretical expectations. Secondly, the loss of experimental data such as code, records, datasets, etc. is also a

risk that can have a huge impact on the overall project schedule. Therefore, I evaluated the risks and took appropriate measures to avoid them. For example, in the beginning, the experiments of AlexNet did not work as expected on several tries, so I optimised the training parameter recording and tuning mechanism by learning from the experience of the Machine Learning module. The optimised training mechanism not only solved the problem of AlexNet, but also helped to train and tune the neural networks later. Having an alternative solution is also a practical risk-avoidance measure. We do not have only one feasible radar gait recognition scheme, but several. The variety of alternatives allows us to select the most reliable and convincing solution through experiments.

Negative Experience

The management of experimental data is another important part of risk management, but I did not perform well in managing experimental data. I have experienced the loss of trained models due to the software error of PyCharm. Since the code is running on a Linux server, PyCharm is used to link the remote Python interpreter and code and data are synchronised between the host and server. The trained models were not backed up immediately after the results were obtained, and a software error caused all data both on the host and the server overwritten by random bits. From this lesson, I began to use Git, which is a version control system that manages iterations of the source code and also backs up the models and results. Git stores the history locally and also stores the backup data on GitHub servers to keep it secure.

5.2.3 Lessons Learned from Conducting Research

The project is research-oriented, so several research processes have been strictly included. With the support and help of my supervisor, progress has been continuously smooth and successful.

The first necessary step was the literature review, which was fundamental to the whole

project as well. The experience from previous research provided me with the basis for conducting a literature review in an unknown field. Radar signals and their different representations were studied under the guidance of my supervisor who is an expert. I used to study only on my own, however, my supervisor has taught me to learn to ask questions. One important lesson I learned from it was that we should not be ashamed to consult with experienced people. It is the same for studying deep learning. What is surprising is that the self-study in deep learning and discussion with fellows has significantly helped me with my understanding of other forms of statistical learning in another module. Just like the transfer learning in machine learning, knowledge can certainly be transferred and be of great help in other fields.

Absorbing new knowledge, especially exposure to a completely new field can be terrifying, but the point here is to take the first step and have the courage to consult experts. After the literature review, reproduction of some valuable existing ideas is conducted to verify their availability, and new ideas are raised in this stage by evaluating the existing ones. Experiments are conducted to test some of the proposed ideas about the methods. According to the experimental results, there are several sprints to evaluate and improve the proposed method. Sprints also helped the project progress smoothly and robustly step by step.

5.3 Conclusion and Future Work

The proposed Attention-based Dual-Stream ViT well solves the problem of radar gait recognition. The radar signal can be represented as a spectrogram or a cadence velocity diagram. The complementary nature of these two representations motivates us to utilise both representations for better classification performance. We propose to use the Vision Transformer to split the spectrogram and CVD into patches and embed their positions so that the information embedded in different frequency bands could be effectively extracted and properly aligned. This enables the model to exploit the deep physical knowledge of

the radar signal. The proposed attention-based fusion scheme integrates the discriminant features from the two representations. The proposed model is compared with state-of-the-art models including AlexNet, VGG and ResNet on our benchmark dataset. We also conducted experiments on the attention-based dual-stream architecture with three backbones. The experimental results demonstrate that the proposed model significantly outperforms all compared methods and confirm the effectiveness of the proposed attention-based dual-stream architecture.

Although the result of this study has already been accepted by the International Conference on Acoustics, Speech and Signal Processing, there are still problems unsolved and aspects that can be improved. In addition to the second semester during which I pay efforts in studying asymmetric attention, in the following time, other possibilities of radar signal representations, neural networks and network architectures will be explored and studied. The power cepstrum represents the rate of change in the spectrum. Conceptually, the cepstrum can be obtained by taking the inverse fast Fourier transform (IFFT) on overlapping windows after taking the logarithm of the magnitude spectrums. This representation has the potential to provide features in another dimension different from the spectrogram and CVD. Swin Transformer addresses the issue that Vision Transformer could not extract very detailed features and is possibly helpful to improve the performance on CVD. Experimental results show the weights for the two streams do not vary much. Asymmetric attention might be able to add prior knowledge to adjust weights and perform better. The results of these studies will hopefully be presented in a journal. As for my personal progress, I have already learned a lot, from rigorous academic writing skills, a complete research cycle, and ideas raising from brainstorming to embracing new knowledge and technologies. These also benefit a lot to my understanding of the domain of computer vision and my future career.

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Appendix A

Full Architecture of ResNet

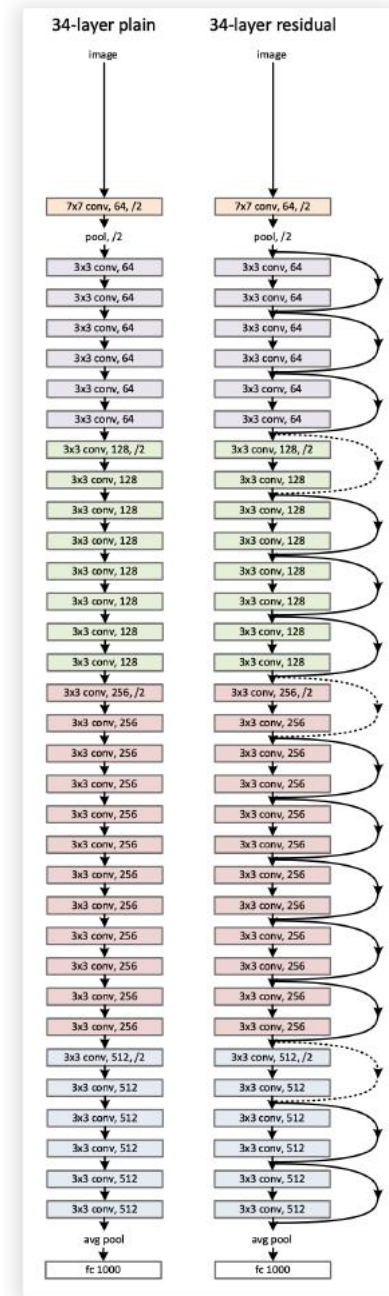


Figure A.1: Architecture of ResNet [6]

Appendix B

Full Experimental Environment Requirements

#	Name	Version	Build	Channel
2	_ipyw_jlab_nb_ext_conf	0.1.0	py37_0	defaults
3	alabaster	0.7.12	py37_0	defaults
4	anaconda	2021.11	py37_0	defaults
5	anaconda-client	1.9.0	py37hecd8cb5_0	defaults
6	anaconda-navigator	2.1.2	py37hecd8cb5_0	defaults
7	anaconda-project	0.10.1	pyhd3eb1b0_0	defaults
8	anyio	2.2.0	py37hecd8cb5_1	defaults
9	appdirs	1.4.4	pyhd3eb1b0_0	defaults
10	applaunchservices	0.2.1	pyhd3eb1b0_0	defaults
11	appnope	0.1.2	py37hecd8cb5_1001	defaults
12	appsript	1.1.2	py37h9ed2024_0	defaults
13	argcomplete	1.12.3	pyhd3eb1b0_0	defaults
14	argh	0.26.2	py37_0	defaults
15	argon2-cffi	20.1.0	py37h9ed2024_1	defaults
16	arrow	0.13.1	py37_0	defaults
17	asn1crypto	1.4.0	py_0	defaults
18	astroid	2.6.6	py37hecd8cb5_0	defaults
19	astropy	4.3.1	py37hf9932de_0	defaults
20	async_generator	1.10	py37h28b3542_0	defaults
21	atomicwrites	1.4.0	py_0	defaults
22	attrs	21.2.0	pyhd3eb1b0_0	defaults
23	autopep8	1.5.7	pyhd3eb1b0_0	defaults
24	babel	2.9.1	pyhd3eb1b0_0	defaults
25	backcall	0.2.0	pyhd3eb1b0_0	defaults
26	backports	1.0	pyhd3eb1b0_2	defaults
27	backports.functools_lru_cache	1.6.4	pyhd3eb1b0_0	defaults
28	backports.shutil_get_terminal_size	1.0.0	pyhd3eb1b0_3	defaults
29	backports.tempfile	1.0	pyhd3eb1b0_1	defaults
30	backports.weakref	1.0.post1	py_1	defaults
31	beautifulsoup4	4.10.0	pyh06a4308_0	defaults
32	binaryornot	0.4.4	pyhd3eb1b0_1	defaults
33	bitarray	2.3.0	py37h9ed2024_1	defaults
34	bkcharts	0.2	py37_0	defaults
35	black	19.10b0	py_0	defaults
36	blas	1.0	mk1	defaults
37	bleach	4.0.0	pyhd3eb1b0_0	defaults
38	blosc	1.21.0	h2842e9f_0	defaults
39	bokeh	2.4.1	py37hecd8cb5_0	defaults
40	boto	2.49.0	py37_0	defaults
41	bottleneck	1.3.2	py37hf1fa96c_1	defaults

42	brotli	1.0.9	hb1e8313_2	defaults
43	brotlipy	0.7.0	py37h9ed2024_1003	defaults
44	brunsl	0.1	h23ab428_0	defaults
45	bzip2	1.0.8	h1de35cc_0	defaults
46	c-ares	1.17.1	h9ed2024_0	defaults
47	ca-certificates	2021.10.26	hecd8cb5_2	defaults
48	cached-property	1.5.2	py_0	defaults
49	certifi	2021.10.8	py37hecd8cb5_0	defaults
50	cffi	1.14.6	py37h2125817_0	defaults
51	cfitsio	3.470	hee0f690_6	defaults
52	chardet	4.0.0	py37hecd8cb5_1003	defaults
53	charls	2.2.0	h23ab428_0	defaults
54	charset-normalizer	2.0.4	pyhd3eb1b0_0	defaults
55	click	8.0.3	pyhd3eb1b0_0	defaults
56	cloudpickle	2.0.0	pyhd3eb1b0_0	defaults
57	clyent	1.2.2	py37_1	defaults
58	colorama	0.4.4	pyhd3eb1b0_0	defaults
59	conda	4.12.0	py37hecd8cb5_0	defaults
60	conda-build	3.21.7	py37hecd8cb5_0	defaults
61	conda-content-trust	0.1.1	pyhd3eb1b0_0	defaults
62	conda-env	2.6.0	1	defaults
63	conda-pack	0.6.0	pyhd3eb1b0_0	defaults
64	conda-package-handling	1.7.3	py37h9ed2024_1	defaults
65	conda-repo-cli	1.0.4	pyhd3eb1b0_0	defaults
66	conda-token	0.3.0	pyhd3eb1b0_0	defaults
67	conda-verify	3.4.2	py_1	defaults
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69	cookiecutter	1.7.2	pyhd3eb1b0_0	defaults
70	cryptography	3.4.8	py37h2fd3fbb_0	defaults
71	curl	7.78.0	h7bc2e8c_0	defaults
72	cycler	0.10.0	py37_0	defaults
73	cython	0.29.24	py37he9d5cce_0	defaults
74	cytoolz	0.11.0	py37haf1e3a3_0	defaults
75	daal4py	2021.3.0	py37h01d92e1_0	defaults
76	dal	2021.3.0	hecd8cb5_555	defaults
77	dask	2021.10.0	pyhd3eb1b0_0	defaults
78	dask-core	2021.10.0	pyhd3eb1b0_0	defaults
79	dataclasses	0.8	pyh6d0b6a4_7	defaults
80	dbus	1.13.18	h18a8e69_0	defaults
81	debugpy	1.4.1	py37h23ab428_0	defaults
82	decorator	5.1.0	pyhd3eb1b0_0	defaults
83	defusedxml	0.7.1	pyhd3eb1b0_0	defaults
84	diff-match-patch	20200713	pyhd3eb1b0_0	defaults
85	distributed	2021.10.0	py37hecd8cb5_0	defaults
86	docutils	0.17.1	py37hecd8cb5_1	defaults
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Appendix C

ICASSP Accepted Paper

ATTENTION-BASED DUAL-STREAM VISION TRANSFORMER FOR RADAR GAIT RECOGNITION

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ABSTRACT

Radar gait recognition is robust to light variations and less infringement on privacy. Previous studies often utilize either spectrograms or cadence velocity diagrams. While the former shows the time-frequency patterns, the latter encodes the repetitive frequency patterns. In this work, a dual-stream neural network with attention-based fusion is proposed to fully aggregate the discriminant information from these two representations. The both streams are designed based on the Vision Transformer, which well captures the gait characteristics embedded in these representations. The proposed method is validated on a large benchmark dataset for radar gait recognition, which shows that it significantly outperforms state-of-the-art solutions.

Index Terms— Radar gait recognition, Spectrogram, Cadence velocity diagram, Vision transformer, Attention-based fusion

1. INTRODUCTION

Human gait recognition has become increasingly attractive in biometric applications such as public safety monitoring, health screening and human-computer interaction [1]. Traditional gait recognition methods [2, 3] often require videos captured from a side view and heavily depend on lighting conditions. Taking pictures of people may cause privacy issues. In contrast, radar can capture the micro-Doppler signatures (mDS) of front-view gait features from a moving target [4], revealing human dynamics and recognize human motions and gestures [4–6]. More importantly, radar has less invasion of privacy and can work robustly in a variety of real-world situations such as dim conditions. In this paper, the problem of front-view human identification using radar mDS is studied.

Many mDS representations have been developed, *e.g.*, spectrograms [7, 8], cadence velocity diagrams (CVD) [9] and cepstrograms [10]. The spectrogram is a time-varying representation of mDS in the time-frequency domain [4], which has been applied to human activity recognition [6, 11].

The time-varying gait information such as the swinging speed of arms and legs can be well encoded in the spectrogram. However, the gait spectrograms of different people have very little differences, while the same person may have different gait characteristics, which makes the human identification using radar gait features particularly challenging. The CVD is another mDS representation by taking Fast Fourier Transform of the spectrogram along the time axis [9]. The CVD is less studied for gait recognition [12]. It provides a useful measurement of repetition patterns for different velocities of body parts, which is a supplement to the spectrogram.

Traditional classifiers such as Naïve Bayes [13], support vector machines [14] and k -nearest-neighbour classifiers [15] have been used to classify mDS, while deep convolutional neural networks (DCNNs) often produce better performance. DCNNs have been used on spectrograms for radar gesture recognition and action classification [4, 6, 16], while other representations such as CVD are not fully explored but could provide complementary information to spectrogram. It is hence advantageous to use both spectrogram and CVD for classification. In addition, those DCNNs directly adopted from image recognition tasks often ignore the unique physical nature of the radar signal compared to optical images.

In this work, an Attention-based Dual-Stream Vision Transformer (ADS-ViT) is proposed to recognize people through radar gaits. As spectrograms often focus on the short-time-varying nature of mDS only, the CVD is introduced to capture the information on how often different frequencies repeat across a long duration. The proposed dual-stream network extracts features from these two representations simultaneously. Both streams are designed based on the Vision Transformer (ViT) to deeply exploit the gait characteristics embedded in patches of spectrogram and CVD. Furthermore, an attention-based fusion network is designed to optimally combine features from two streams. The proposed framework is validated on a large dataset for radar gait recognition, which shows that it significantly outperforms state-of-the-art models.

Our contributions are two-fold: 1) The proposed ADS-ViT effectively extracts and fuses features from spectrogram and CVD for radar gait recognition. 2) Both streams of ADS-ViT utilize the patch-processing ability of ViT to effectively

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capture the gait characteristics embedded in patches corresponding to frequency bands of spectrogram and CVD.

2. PROPOSED ATTENTION-BASED DUAL-STREAM VISION TRANSFORMER

2.1. Overview of Proposed Method

The proposed ADS-ViT for radar gait recognition is shown in Fig. 1(a). To better exploit the discriminant information embedded in the radar signal, two initial feature representations, spectrogram [7, 8] and CVD [9], are jointly utilized in the proposed framework. Two subnetworks, spectrogram stream and CVD stream, are used to extract features from each of these two representations, respectively. For spectrograms and CVDs, different image patches correspond to different frequency bands. To robustly extract low-level features from these frequency bands, a network based on the Vision Transformer (ViT) [17] is designed, where the local features are encoded through the Position and Patch Embedding mechanism of ViT. ViT could also capture the high-level semantic information across different frequency bands by utilizing the self-attention mechanism in the Transformer Encoder of ViT. An attention-based fusion mechanism is proposed to find the most discriminant features, and fuse the features of two streams into one feature map. Finally, a fully-connected layer is used to predict the person's identity given the radar signal.

More specifically, the proposed ADS-ViT network can be formulated as a quadruplet $\mathcal{Q} = (\mathcal{V}_s, \mathcal{V}_c, \mathcal{F}, \mathcal{C})$, where $\mathcal{V}_s$ and $\mathcal{V}_c$ are the ViT networks for spectrogram and CVD respectively, as shown in Fig. 1(b), $\mathcal{F}$ is an attention-based fusion network and $\mathcal{C}$ is the classifier at the end. Denote the generated spectrogram and CVD as $\mathbf{I}_s$ and $\mathbf{I}_c$, respectively. The features $\mathbf{f}_s$ from the spectrogram stream and $\mathbf{f}_c$ from the CVD stream are generated as:

$$\mathbf{f}_s = \mathcal{V}_s(\mathbf{I}_s), \quad (1)$$

$$\mathbf{f}_c = \mathcal{V}_c(\mathbf{I}_c), \quad (2)$$

These two sets of features are fused by the attention-based fusion network $\mathcal{F}$ to generate the features $\mathbf{f}_F$ as:

$$\mathbf{f}_F = \mathcal{F}(\mathbf{f}_s, \mathbf{f}_c). \quad (3)$$

Finally, the label $\hat{l}$ is predicted by the classifier $\mathcal{C}$ as:

$$\hat{l} = \mathcal{C}(\mathbf{f}_F). \quad (4)$$

2.2. Time-frequency Representations

Spectrograms [7, 8] and CVDs [9] are jointly utilized for robust radar gait recognition in this paper.

Spectrogram: Tahmoush and Silvius [12] modeled Doppler of each body part of a walking human as sinusoidal modulation in the spectrogram. This model reveals the velocities of

each body part and determines the necessary human gait characteristics. The extracted features are the mean Doppler velocity and the size of torsos in the spectrogram. The mDS of a person is found relatively consistent and different walking people exhibit discriminative characteristics in the spectrogram [12]. The radar spectrogram can well manifest the time-varying characteristics of a person's gait. A sample spectrogram is shown in Fig. 1(a).

Cadence velocity diagram: After deriving the spectrogram, the CVD [9] is obtained by taking the Fourier transform of the spectrogram along the time axis. The derived CVD, as shown in Fig. 1(a), is a matrix with rows representing Doppler frequencies and columns representing cadence frequencies, which measures how frequently different frequencies appear in the signal over the observation duration. It encodes useful information of the repetition of velocities, which would be the key feature to identify a walking person.

2.3. Vision Transformer for Spectrogram and CVD

Two network branches based on the Vision Transformer are designed to extract features from spectrograms and CVDs, respectively. The ViT network consists of two parts: Position and Patch Embedding (PPE) and Transformer Encoder (TE), as shown in Fig. 1(b). The original size of spectrogram/CVD is 115×115 . To fit the pretrained ViT network, the input spectrogram/CVD is converted to a 3-channel image $\mathbf{I} \in \mathcal{R}^{224 \times 224 \times 3}$, by replicating the resized grayscale image for three channels. In the PPE stage, $\mathbf{I}$ is convoluted into the feature map of size $14 \times 14 \times 768$ via a 16×16 kernel with a stride of 16, and flattened to the features $\mathbf{f}_I \in \mathcal{R}^{196 \times 768}$. Similarly as in [17], the learnable classification token $\mathbf{t} \in \mathcal{R}^{1 \times 768}$ is prepended to $\mathbf{f}_I$ as:

$$\mathbf{f}'_I = [\mathbf{t}, \mathbf{f}_I]. \quad (5)$$

The position information is embedded by adding a learnable position matrix $\mathbf{P} \in \mathcal{R}^{197 \times 768}$ for retaining positional knowledge of each patch, similarly as in [17], which results in the feature map $\mathbf{f}_P \in \mathcal{R}^{197 \times 768}$ with position embedding to the Transformer Encoder as:

$$\mathbf{f}_P = \mathbf{P} + \mathbf{f}'_I. \quad (6)$$

The TE module consists of 12 repeating blocks, where each contains a multi-head self-attention layer. This module could learn the global relations between patches. The learned knowledge is represented in the classification tokens for all patches and used as the output of the ViT network.

Unlike real-world images in which an object could be positioned anywhere in an image but interpreted the same, patches of spectrogram or CVD generated from the radar signal have the unique physical meanings when positioned at different locations in the image. Each patch in spectrogram or CVD contains the information in different frequency bands

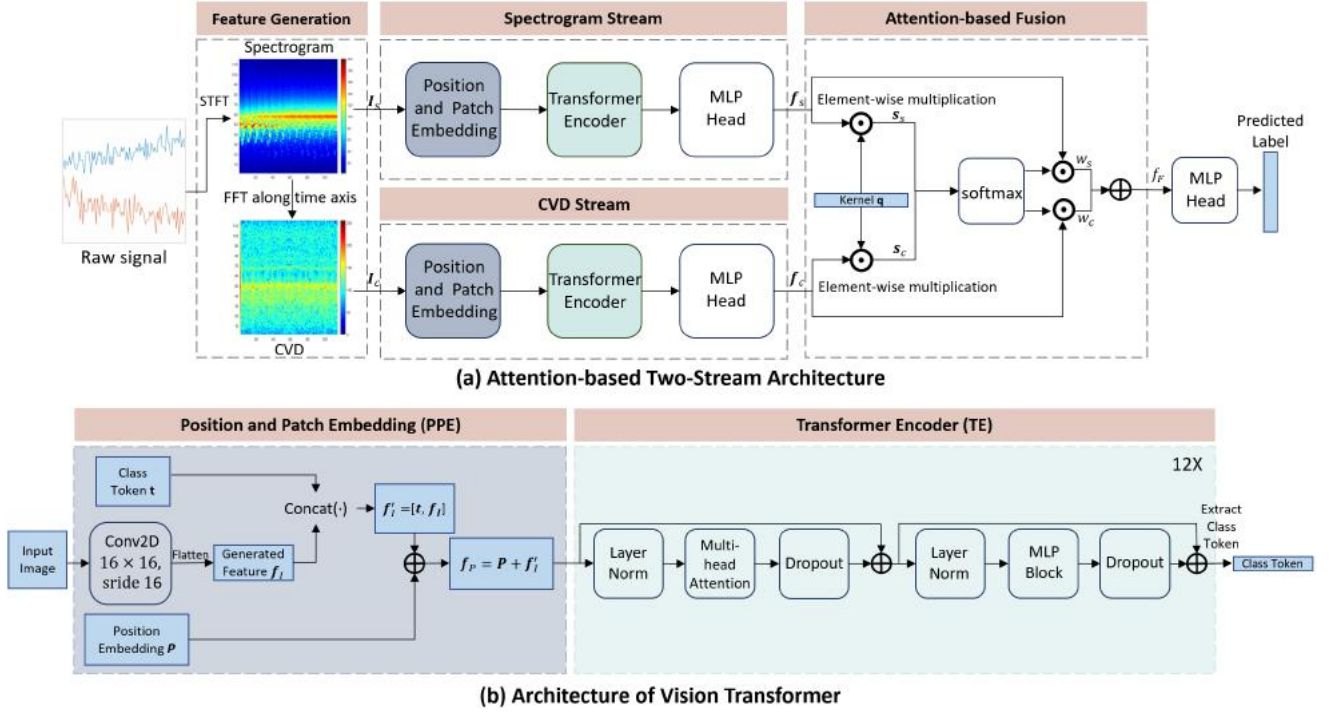


Fig. 1. Overview of the proposed method shown in (a) and pipeline of ViT shown in (b).

at different time instances (or different cadence velocities for CVD). Even patches similar in appearance could be physically different based on their frequency bands and temporal positions.

The commonly-used convolutional neural networks such as AlexNet [16], VGG [4] and ResNet [4] often use the same convolutional kernel across the whole image, while the proposed ADS-ViT treats patches differently, *i.e.*, it splits the spectrogram/CVD into patches and embeds their positions so that the information embedded in different frequency bands could be properly aligned and effectively extracted. This partially justifies the superior performance of the proposed ADS-ViT over other models. The multi-head self-attention mechanism in the Transformer Encoder could further extract the reserved discriminant information from each patch and process this information globally to encode the global relations among patches.

2.4. Attention-based Fusion

Inspired by Chen *et al.* [18], an attention-based fusion architecture shown in Fig. 1(a) is developed to fuse the complementary features extracted from both spectrogram and CVD. The target of the attention-based feature-level fusion is to find a set of weights $\{w_i \in \mathcal{R}^{1 \times 768}\}$ for the features

$\{f_i \in \mathcal{R}^{1 \times 768}\}$ to obtain an aggregated feature $f_a \in \mathcal{R}^{1 \times 768}$:

$$f_a = \sum_{i=1}^n w_i \odot f_i, \quad (7)$$

where $\odot$ denotes element-wise multiplication, n is the number of features and $n = 2$ in this paper. In the attention-based fusion model, the kernel $q \in \mathcal{R}^{1 \times 768}$ is required to be trained. The process begins with the element-wise multiplication between the feature vector f_i and the kernel q :

$$s_i = q \odot f_i, \quad (8)$$

where $s_i \in \mathcal{R}^{1 \times 768}$ are the confidence scores for feature representations. A softmax function is then applied to s_i to assure that the derived weights $\sum_i w_i = \mathbf{1} \in \mathcal{R}^{1 \times 768}$:

$$w_i = e^{s_i} \oslash \sum_j e^{s_j}, \quad (9)$$

where $\oslash$ denotes element-wise division. The fused feature vector is finally generated using Eq. (7). This attention-based fusion could well highlight the most discriminant features by training the kernel function q .

3. EXPERIMENTAL RESULTS

3.1. Dataset

There is no publicly available dataset for radar gait recognition, and hence a K-MC1 radar transceiver and an ST200 eval-

uation system launched by RFbeam Microwave GmbH are employed to collect a radar gait dataset by the authors. There are two data collection sessions at least two weeks apart. In each session, each volunteer walks 10 sequences. (Some only participate in the first session.) The dataset consists of 1670 walking sequences from 98 volunteers. For each volunteer, 50% of sequences are randomly selected as the training set and the rest are used for testing. In each sequence, a volunteer walks away from the radar along a corridor about 40 meters long, turns around and walks back towards the radar, lasting about 30 seconds. The sampling rate of the original signal is 125k. After a decimation of 64, the sampling rate is about 1.95k. The spectrogram is built using 128 sample points with an overlapping ratio of 90%. After removing the cluster and non-informative high-frequency components, the spectrum has 115 data points. To enrich the dataset, each sequence is cut into multiple frames of 115 data points, with a stride of 10. As a result, a total number of 45,768 frames of size 115×115 are generated, in which 22,894 are used for training and 22,874 for testing.

3.2. Experimental Settings

AlexNet has been used as the backbone of the Deep Convolutional Neural Network on spectrograms of the radar signal to identify people [16]. In [4], VGG16 [19] and ResNet18 [20] have been used for radar gait recognition. These three approaches are implemented and evaluated on our benchmark dataset. As the proposed ADS-ViT utilizes both spectrogram and CVD, AlexNet [16], VGG16 [4] and ResNet18 [4] are applied on the CVD for comparison as well.

The size of frames is 115×115 initially and resized to 224×224 to fit the networks. Stochastic Gradient Descent is used as the optimizer of all models with a momentum of 0.9 and a weight decay of 5×10^{-5} . The cross-entropy loss function is used. For learning parameters, a linear learning rate warmup and decay is used with a learning rate scheduler, and the initial learning rate is set to 0.01 for ResNet, 0.001 for AlexNet and 0.0001 for VGG. The batch size is 128 for all models. The proposed ADS-ViT initialize ViT networks with pretrained weights for both spectrogram and CVD streams. All the models are trained for 500 epochs and the optimal performance on the test set is reported.

3.3. Comparisons to State-of-the-art Approaches

The comparisons to state-of-the-art models are summarized in Table 1. Most existing models are applied on spectrograms, *e.g.*, AlexNet [16], VGG16 [4] and ResNet18 [4]. As the proposed method utilizes both spectrogram and CVD, these three CNN models are applied on the CVD as well.

Table 1 shows that the proposed ADS-ViT significantly outperforms all compared methods. Compared to the second-best model, ResNet18 on spectrogram [4], the proposed

Table 1. Comparisons to state-of-the-art AlexNet [16], VGG16 [4] and ResNet18 [4]. These models are implemented and evaluated on both spectrograms and CVDs on our benchmark dataset for comparison.

Method	Accuracy
DCNN-AlexNet on Spectrogram [16]	71.56%
VGG16 on Spectrogram [4]	69.24%
ResNet18 on Spectrogram [4]	85.56%
DCNN-AlexNet on CVD	72.44%
VGG16 on CVD	79.83%
ResNet18 on CVD	80.73%
Proposed ADS-ViT	91.02%

method improves the classification accuracy from 85.56% to 91.02%. The achieved significant performance gain is attributed to the proposed ADS-ViT in two folds. First, by dividing the spectrogram/CVD into patches, the proposed ViT network extracts the local features from different frequency bands using the Position and Patch Embedding mechanism and extracts the global features across different frequency bands using the self-attention mechanism embedded in the transformer encoder.

Second, while the spectrogram encodes the short-duration information using the short-time Fourier transform, the long-duration repeating frequency patterns are captured by the CVD. By utilizing both spectrogram and CVD, the short-duration repetitive patterns in the spectrogram and the long-duration repetitive frequency patterns in the CVD are captured by the two streams of the proposed ADS-ViT, respectively. The proposed attention-based fusion scheme well integrates the discriminant information extracted from spectrogram and CVD, which leads to the superior performance of the proposed ADS-ViT.

4. CONCLUSION

The proposed Attention-based Dual-Stream ViT well solves the problem of radar gait recognition. The radar signal can be represented as a spectrogram or a cadence velocity diagram. The complementary nature of these two representations motivates us to utilize both representations for better classification performance. We propose to use the Vision Transformer to split the spectrogram and CVD into patches and embed their positions so that the information embedded in different frequency bands could be effectively extracted and properly aligned. This enables the model to exploit the deep physical knowledge of the radar signal. The proposed attention-based fusion scheme integrates the discriminant features from the two representations. The proposed model is compared with state-of-the-art models on our benchmark dataset. The experimental results demonstrate that the proposed model significantly outperforms all compared methods.

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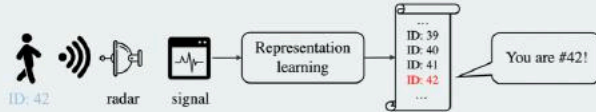
Appendix D

ICASSP Poster

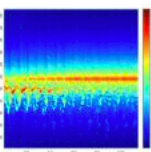
Attention-based Dual-stream Vision Transformer for Radar Gait Recognition

01 Motivations

Side-view human gait recognition relies heavily on light conditions and may cause infringement on privacy.



Radar can capture the micro-Doppler signatures (mDS) of front-view gait features from a moving target.

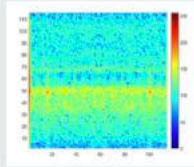


Spectrogram:

A time-varying representation of mDS in the joint time-frequency domain.

Cadence Velocity Diagram (CVD):

Provide another useful measure on the frequency of different velocities of body parts repeat.



02 Contributions

The proposed attention-based dual-stream Vision Transformer (ADS-ViT) effectively extracts and fuses features from spectrogram and CVD for radar gait recognition.

Patch-processing ability of ViT effectively captures the gait characteristics embedded in patches.

03 Methodology

The raw radar signal is first processed using STFT to form the spectrograms.

CVDs are converted from the spectrograms by applying fast Fourier transform along time axis.

$$f_s = \mathcal{V}_s(I_s),$$

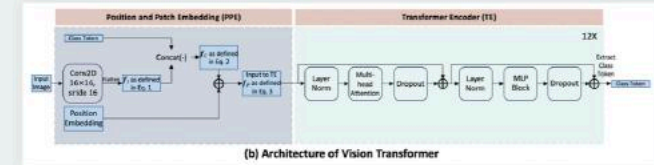
$$f_c = \mathcal{V}_c(I_c),$$

Two streams of Vision Transformer are utilised for extracting features from two representations.

An attention-based feature level fusion module aggregates discriminant features from two streams.

$$f_F = \mathcal{F}(f_s, f_c).$$

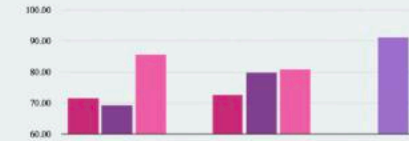
Feature extractor: Vision Transformer (ViT)



In spectral representations, the spatial coordinates correspond to unique physical interpretations.

ViT can deeply exploit the gait characteristics embedded in patches of spectrogram and CVD.

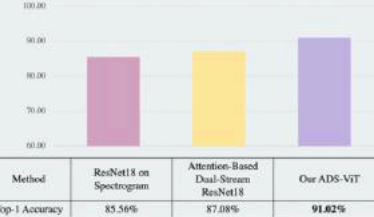
04 Experimental Results



	Spectrogram	CVD
DCNN-AlexNet	71.56%	72.44%
VGG16	69.24%	79.83%
ResNet18	85.56%	80.73%
Ours		91.02%

The proposed ADS-ViT significantly outperforms all compared methods with an accuracy of 91.02%

Ablation Study



Ablation study:

ADS architecture improves performance of ResNet18.

From

85.56%

To

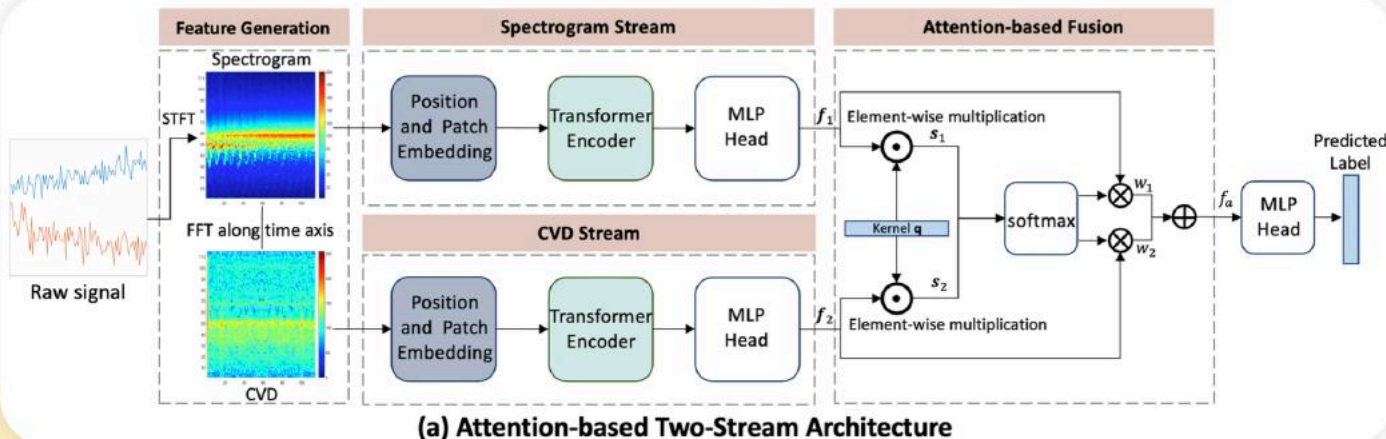
87.08%

05 Conclusion

The ADS-ViT well solves the radar gait recognition

Complementary features of spectrogram and CVD are fused

ViT deeply exploits embedded information using patches



Appendix E

ICASSP Presentation Slides

Attention-based Dual-stream Vision Transformer for Radar Gait Recognition

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¹University of Nottingham, Ningbo China

²Nanyang Technological University, Singapore

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Outline

- I. Introduction
- II. The proposed Attention-based Dual-Stream Vision Transformer (ADS-ViT)
- III. Experiments and Results
- IV. Conclusion

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Outline

I. Introduction

II. The proposed Attention-based Dual-Stream Vision Transformer (ADS-ViT)

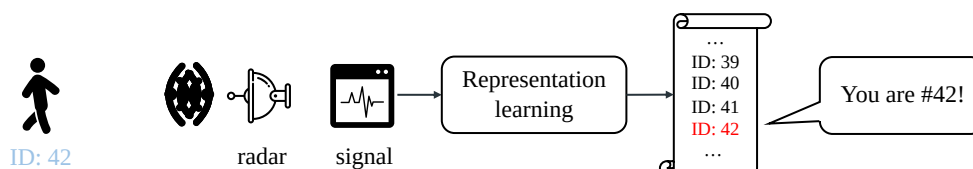
III. Experiments and Results

IV. Conclusion

3

Introduction

- Side-view human gait recognition relies heavily on light conditions and may cause infringement on privacy.
- Radar can capture the micro-Doppler signatures (mDS) of front-view gait features from a moving target.



4

Introduction

icnssp 2022
Singapore

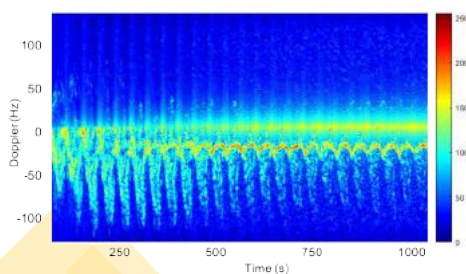


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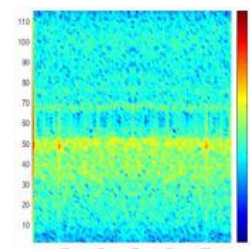
Introduction

icnssp 2022
Singapore

- The spectrogram is a time-varying representation of mDS in the joint time-frequency domain.
- The Cadence Velocity Diagram (CVD) can be converted from spectrogram and provides another useful measure on the frequency of different velocities of body parts repeat.



Spectrogram



CVD

6

Contribution



- The proposed ADS-ViT effectively extracts and fuses features from spectrogram and CVD for radar gait recognition.
- Both streams of ADS-ViT utilise the patch-processing ability of ViT to effectively capture the gait characteristics embedded in patches corresponding to frequency bands of spectrogram and CVD.

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Outline



I. Introduction

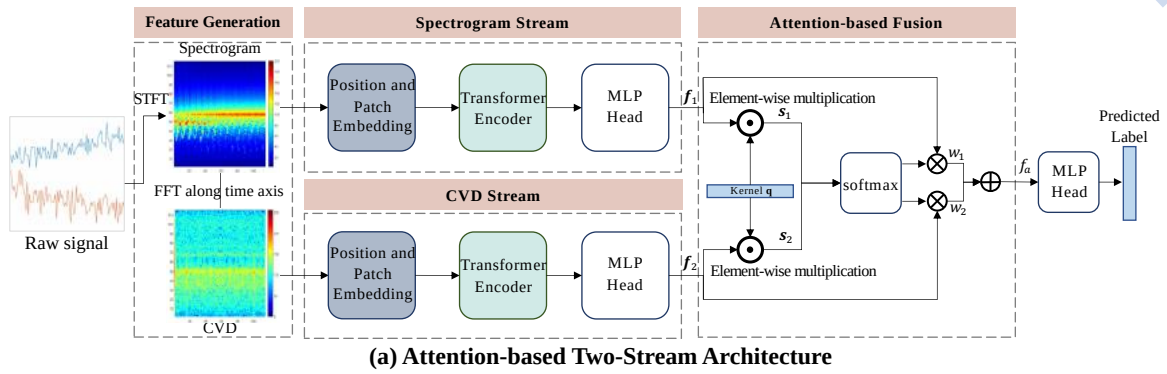
II. The proposed Attention-based Dual-Stream Vision Transformer (ADS-ViT)

III. Experiments and Results

IV. Conclusion

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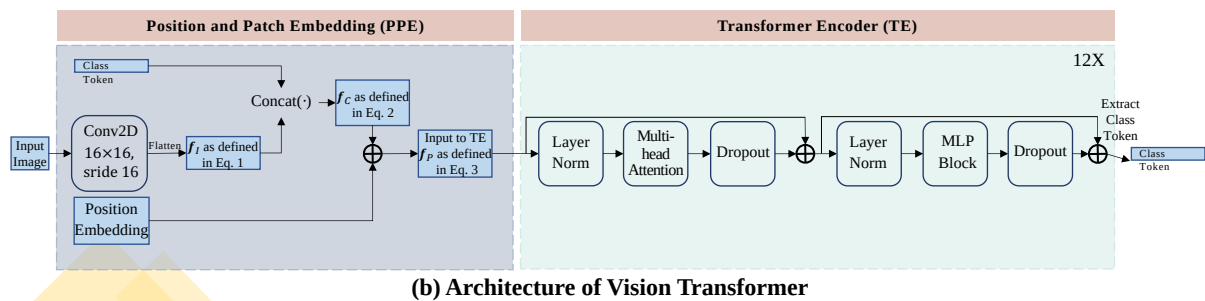
Methodology



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Methodology

- Feature extractor: Vision Transformer (ViT)
- In spectral representations, the spatial coordinates correspond to unique physical interpretations.
- ViT can deeply exploit the gait characteristics embedded in patches of spectrogram and CVD.



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Methodology



- The time-varying gait information such as swinging speed of arms and legs can be well encoded in the spectrogram.
- The CVD provides a useful measurement of repetition patterns for different velocities of body parts, which is a supplement to spectrogram.
- ViT exploits the physical latent of the two representations.
- Fusion mechanism - Attention-based feature-level fusion:

$$f_a = w_s \odot f_s + w_c \odot f_c$$

- Instead of learning w , a kernel q is trained.

$$d_i = q \odot f_i, \quad i \in \{s, c\}$$

$$w_i = \sigma(d_i) = e^{d_i} \oslash \sum_j e^{d_j}, \quad i \in \{s, c\}$$

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Outline



I. Introduction

II. The proposed Attention-based Dual-Stream Vision Transformer (ADS-ViT)

III. Experiments and Results

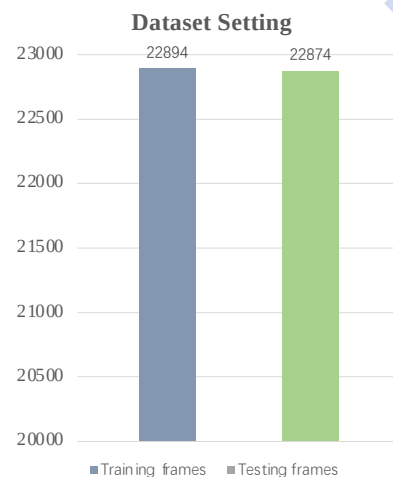
IV. Conclusion

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Experiments

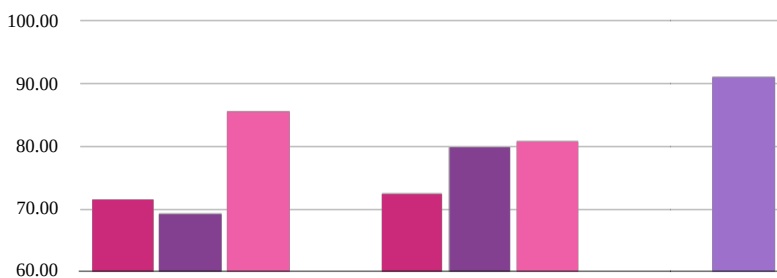
Dataset

- A K-MC1 radar transceiver and an ST200 evaluation system launched by RFbeam Microwave GmbH are employed to collect a radar gait dataset by the authors
- 1670 walking sequences from 98 volunteers
- 22,894 frames are used for training and 22,874 for testing



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Results

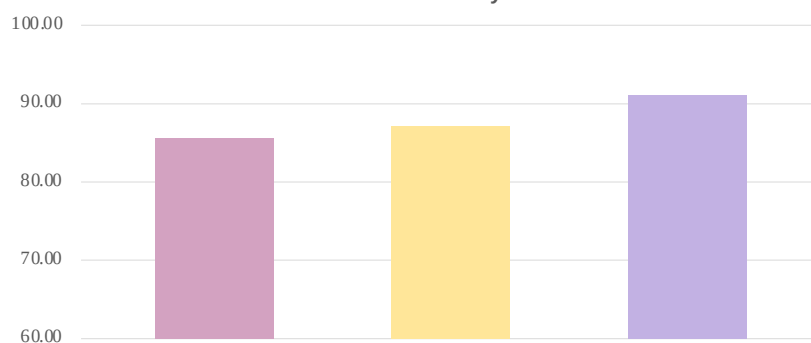


	Spectrogram	CVD
DCNN-AlexNet	71.56%	72.44%
VGG16	69.24%	79.83%
ResNet18	85.56%	80.73%
Ours		91.02%

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Results

Ablation Study



Method	ResNet18 on Spectrogram	Attention-Based Dual-Stream ResNet18	Our ADS-ViT
Top-1 Accuracy	85.56%	87.08%	91.02%

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Outline

I. Introduction

II. The proposed Attention-based Dual-Stream Vision Transformer (ADS-ViT)

III. Experiments and Results

IV. Conclusion

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Conclusion



- Contributions
 - The proposed ADS-ViT effectively extracts and fuses features from spectrogram and CVD for radar gait recognition.
 - Both streams of ADS-ViT utilize the patch-processing ability of ViT to effectively capture the gait characteristics embedded in patches corresponding to frequency bands of spectrogram and CVD.
- Experiments
 - Outperforms AlexNet, VGG and ResNet.
 - Perform the best in ablation study.

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Thank you!



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Appendix F

Training Record of Attention-based Dual-stream AlexNet

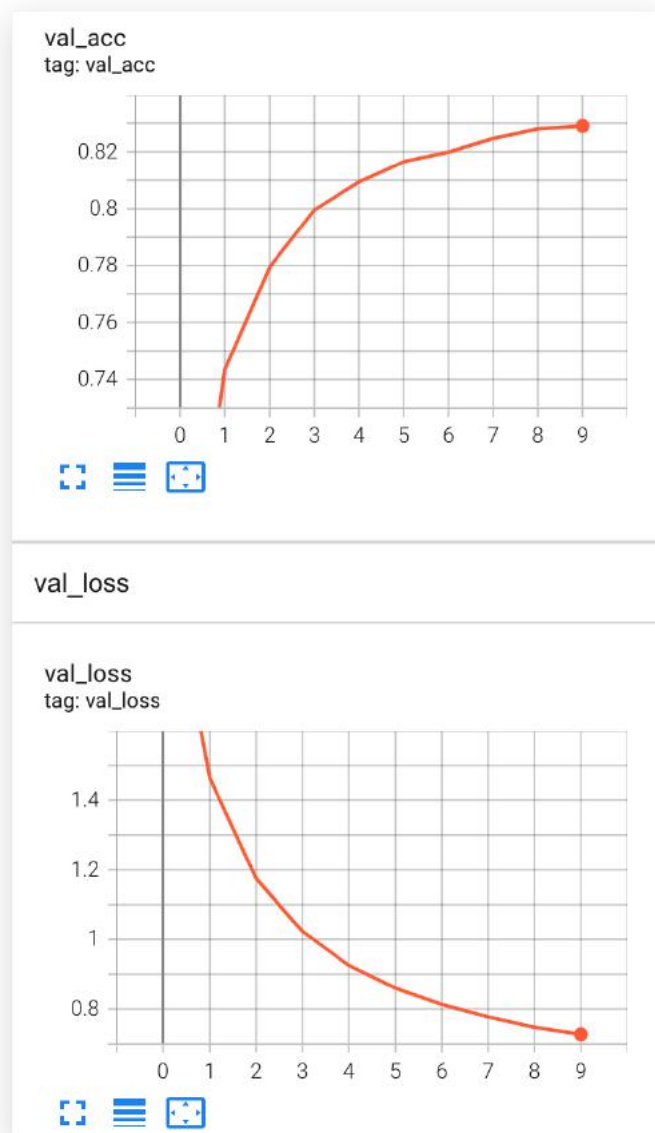


Figure F.1: Training Record of Attention-based Two-stream AlexNet

Appendix G

Training Record of Attention-based Dual-stream VGG

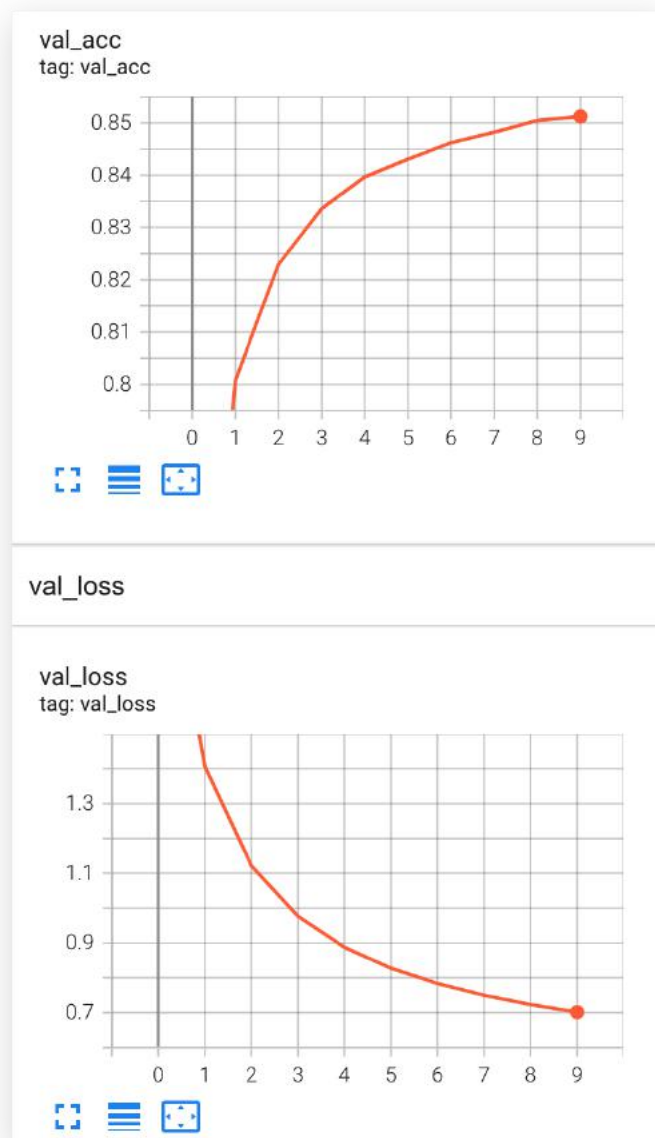


Figure G.1: Training Record of Attention-based Two-stream VGG

Appendix H

Training Record of Attention-based Dual-stream Vision Transformer

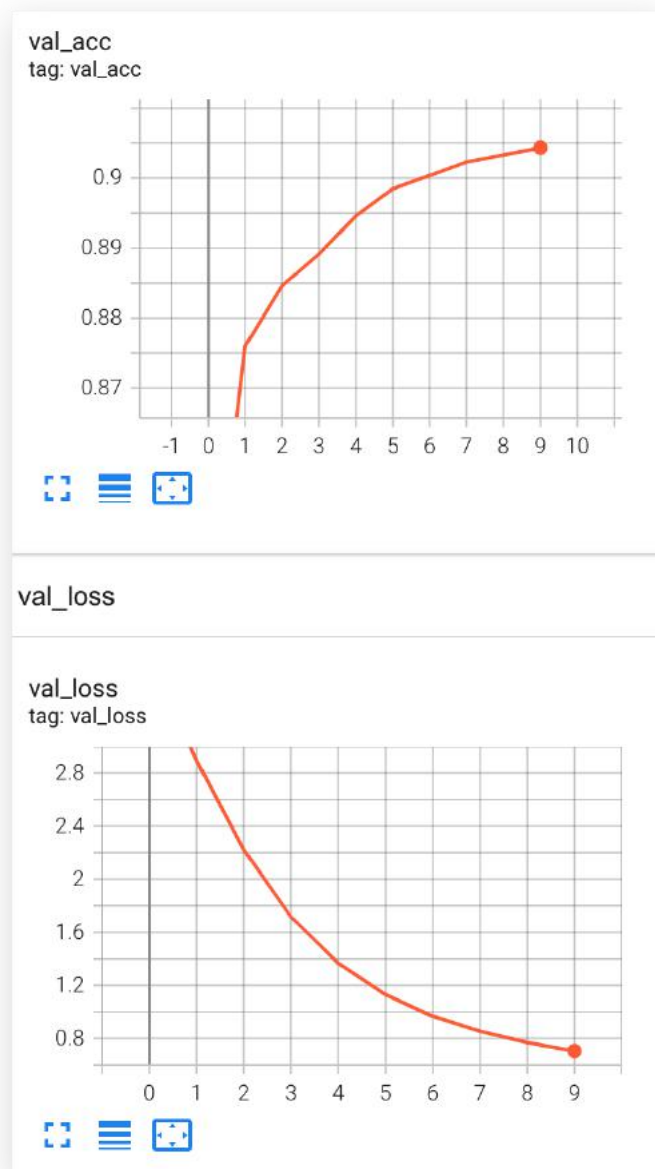


Figure H.1: Training Record of Attention-based Two-stream Vision Transformer